MILITARY SPECIFICATION

RECEIVER, RADIO, R-1051 ( )/URR

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Radio Receiver, R-1051( )/URR and associated accessories, hereinafter referred to as the equipment. The equipment is a digitally tuned, super-heterodyne receiver. The equipment operates in the 2.0 megahertz (MHz) to 30.0 MHz frequency range and provides reception of single sideband (SSB), selectable upper sideband (USB), lower sideband (LSB), and independent sideband (ISB), continuous wave (CW), amplitude modulation (AM) (compatible), single and multi-channel teletypewriter and Naval Tactical Data System (NTDS) Link 11 signals.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-R-94	Resistors, Variable, Composition, General Specification For
MIL-S-901	Shock Tests, H.I. (High-Impact); Machinery, Equipment And Systems, Requirements For
MIL-C-915	Cable And Cord, Electrical, For Shipboard Use, General Specification For
MIL-M-10304	Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification For
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-W-16878/4	Wire, Electrical, Polytetrofluoroethylene (PTFE) Insulated, 200 Deg. C, 600 Volts, Extruded Insulation
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts): Packaging Of
MIL-F-19207	Fuseholders, Extractor Post Type, Blown Fuse Indicating And Nonindicating, General Specification For
MIL-C-24056	Cabinets, Electronic Equipment, CY-4516(A)/S And CY-4516(B)/S (Single Bay) (Naval Shipboard Use)
MIL-M-38510	Microcircuits, General Specification For
MIL-C-39012	Connectors, Coaxial, Radio Frequency, General Specification For

Benefical comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-8111), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5820

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection by Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-188-203-1	Subsystem Design And Engineering Standards For Tactical Digital Information Link (TADIL A)
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810C	Environmental Test Methods
MIL-STD-965	Parts Control Program
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

2.1.2 Other Government drawings and publication. The following other Government drawings and publication form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

MILITARY

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

RE-B2683892	Installation Control Drawings, Data List, For MK-979/URR
RE-B5034104	Installation Control Drawings, Data List, For R-1051( )/URR

(Copies of specifications, standards, drawings, and publication, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA RS-310-C-77	Racks, Panels And Associated Equipment

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General. The equipment shall be in accordance with MIL-E-16400, to the extent specified herein.

3.1.1 First article. When specified (see 6.2.1), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3). The sample(s) shall be suitable for complete evaluation of mechanical and electrical form, design, and performance. The sample(s) shall be of final mechanical and electrical form, employ standard or nonstandard (when approved by the procuring activity) parts, and be completely representative of final production equipment.

3.1.2 Parts, materials, and processes. Parts, materials, and processes shall be in accordance with the Parts, materials, and processes paragraph of MIL-E-16400, except as specified in 3.1.2.1 through 3.1.2.7. Particular attention shall be given to the acceptable and unacceptable materials of MIL-E-16400.

3.1.2.1 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

3.1.2.2 Prior parts approval. The equipment furnished in accordance with this specification shall employ standard parts or approved nonstandard parts. Nonstandard part approvals previously granted under earlier contracts for equipments represented by any Government-furnished equipment (GFE) shall not be construed as being extended to equipments furnished in accordance with this specification. Nonstandard parts cited in Government-furnished drawings (DWGs) shall not be construed as being approved for use in equipments furnished in accordance with this specification.

3.1.2.3 Individual selection. The performance of the equipment shall not be dependent on the selection of individual parts, unless such selection is specifically approved by the procuring activity.

3.1.2.4 Electron tubes. All electron tubes shall be selected from among approved types in accordance with the Tubes, electron paragraph of MIL-E-16400. Electron tubes shall be limited to applications corresponding to circuits of the GFE.

3.1.2.5 Microelectronic devices. Selection and application of microelectronic devices shall conform to MIL-STD-454, Requirement 64, requiring MIL-M-38510, Class B devices. When deviations from these requirements are necessitated due to availability or cost considerations, hermetically sealed devices, inspected and tested in accordance with the Other microcircuits paragraph of MIL-STD-454, Requirement 64, shall be selected.

3.1.2.6 Semiconductor devices. All semiconductor devices shall be made of silicon.

3.1.2.7 Type BNC and Type N radio frequency (RF) connectors. Where Type BNC and Type N RF coaxial connectors are specified herein, connectors shall be selected in accordance with MIL-C-39012.

3.1.3 Production equipment. Production equipment shall be identical to the equipment that has been subjected to, and passed, the first article examinations and tests (see 3.1.1 and 4.3).

3.1.4 Operational requirements. Each unit shall give reliable performance within the limits specified herein, under conditions of intermittent or continuous operation during exposure to the adverse conditions normally encountered in Naval ship and shore communications service. Mechanical and electrical ruggedness is of paramount importance for assurance of reliable performance without failure.

3.2 GFE. GFE shall be as specified (see 6.2.1 and 6.4).

3.3 Interchangeability. Interchangeability requirements shall be as specified (see 6.2.1).

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3.4 Composition. The equipment shall consist of one each of the items specified in a and b:

a. Receiver, R-1051( )/URR
b. Set of mating connectors, including cable clamps, for equipment system connection, as specified in 1 through 7:
1. USB audio
2. LSB audio
3. Alternating current (AC) power
4. Antenna input
5. Frequency standard input
6. Frequency standard output
7. Auxiliary output (J4)

3.4.1 Accessories. In addition to the basic equipment specified in 3.4, the accessory items specified in a and b shall be furnished in the quantities specified (see 6.2.1):

a. Equipment plug-in assembly extension cables (see 3.15.1.35.1), as specified in 1 through 5:
1. RF amplifier assembly cables (two each)
2. Frequency standard assembly cable (one each)
3. Translator and synthesizer assembly cables (two each)
4. Receiver intermediate frequency (IF) audio cable (one each)
5. Receiver mode selector assembly cables (two each)
b. Set of printed circuit board (PCB) extender cards, one for each plug-in PCB in the equipment except for A2A12 (see 3.15.1.35.2)
c. Shock and vibration mounting assembly (see 3.8.1)

3.5 Frequency range. The frequency (tuning) range of the equipment shall be from 2.0 MHz to 30.0 MHz. Each equipment shall be capable of adjustment to any frequency from 2.0 MHz to 29.9999 MHz in 100-hertz (Hz) steps. In addition, continuous coverage shall be provided by vernier frequency control capable of tuning between 1 kilohertz (kHz) intervals (see 3.16.10.2).

3.5.1 Method of tuning. Frequency selection over the complete frequency range of the equipment shall be accomplished in steps of 100 Hz, where changing the setting of the digital tuning knobs is all that is required to tune the equipment (see 3.15.1.6).

3.6 Primary power requirements. The equipment shall be designed to operate from electrical power having characteristics as specified for Type I of DoD-STD-1399, Section 300, and as specified herein. Each equipment shall be capable of operating and providing rated performance from a primary power source of 115 volts alternating current (VAC) ±10 percent, 48 Hz to 420 Hz, single-phase, with no degradation of performance by power line voltage variations (relative to the nominal) of ±5 percent or above specified power line frequency variations, with either of these variations acting separately or in combination. The total power consumption at 115 volts (V), 48 Hz to 420 Hz, shall not exceed 70 watts.

3.6.1 Transient protection. The equipment shall operate without damage or permanent change in characteristics due to the transient voltage, transient frequency, and power interruption tests of MIL-E-16400 and the 2500 V spike test of DoD-STD-1399, Section 300. Protection shall be included against damage due to transients produced by operation of relays or tuning motors within the equipment.

3.6.2 Electrical overload protection. Electrical overload protection shall be in accordance with the Electrical overload protection paragraph of MIL-E-16400, except that each line of the AC power input shall be protected by fuses.

3.6.3 Leakage current. The equipment leakage current measured in accordance with the procedure of 4.5.4 shall not exceed 5 milliamperes AC or direct current (DC).

3.7 Operational requirements. Operational requirements shall be as specified in 3.7.1 and 3.7.2.

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3.7.1 Type of emission. Each equipment shall provide for the reception of the types of emissions specified in a through l:

a. A1 telegraphy without the use of a modulating audio frequency (CW)
b. A2 telegraphy by the ON-OFF keying of an amplitude-modulating audio frequency or audio frequencies, or by the ON-OFF keying of the modulating emission modulated continuous wave (MCW)
c. A3 telegraphy, double sideband (DSB), full carrier
d. A3a telephony, DSB, full carrier
e. A3b telephony, SSB, reduced carrier
f. A3h telephony, SSB full carrier
g. A3j telephony, SSB suppressed carrier
h. A9b two sidebands; combination of telephony and telegraphy (ISB and single-channel telegraphy)
i. F2j single-channel voice frequency telegraphy radio teletype (RATT)
j. F7j multi-channel voice frequency telegraphy RATT
k. F9 differentially coherent phase shift keying (DPSK), Tactical Digital Information Links (TADIL) A, as specified in JCS 10, DSB
l. F9j differentially coherent DPSK; TADIL A, as specified in MIL-STD-188-203-1, SSB

3.7.2 Modes of operation. The equipment shall be capable of receiving the emissions specified in 3.7.1 by selection of one of the modes of operation specified in a through f:

a. LSB
b. USB
c. ISB
d. AM
e. CW telegraphy
f. Single channel telegraphy, RATT

3.8 Equipment arrangement. Normal installation arrangements for the equipment shall include the arrangements specified in a through c:

a. Mounting using a MT-3114( )/UR shock and vibration mounting assembly
b. Multiple CY-4516(A)/, or equivalent, hard-mounted installation in shipboard electrical equipment cabinets
c. Mounting in an EIA RS-310-C-77 mounting rack

3.8.1 Shock and vibration mounting assembly, MT-3114( )/UR. Each MT-3114( )/UR shall be delivered with the necessary brackets and hardware required to attach the equipment to the mounting assembly (3.15.3).

3.9 Radio receiver system. The contractor shall be responsible for demonstrating complete system operation for the equipment to be delivered in accordance with the requirements of this specification. Equipment provisions for external cabling to the standard Navy shipboard radio remote control system shall be required. Standard Navy cables shall be as specified in MIL-C-915. Mating connectors for all external cabling which requires manufacture by the installing activity shall be furnished with each equipment (see 3.4b). Additional information pertaining to a typical installation of the equipment shall be as specified in DWG RE-B2683892 and DWG RE-B5034104.

3.10 Dimensions and weights. The maximum dimensions and weights of the equipment shall not exceed those shown in TABLE I.

3.11 Environmental service conditions. The equipment shall be capable of continuous, reliable operation, within the performance limits specified in 3.16, under the environmental conditions of MIL-E-16400 specified in a through g and 3.16. The equipment shall operate within tolerances throughout the corresponding test specified in Section 4 without alignment or adjustment, other than external controls normally employed for operation of the equipment.

a. Operating temperature range (Range 4)
b. Nonoperating temperature range (Range 4)
c. Humidity (≤95 percent)
d. Salt fog (spray) (sheltered)
e. Shock (mounted on MT-3114( )/UR)
f. Vibration (mounted on MT-3114( )/UR)
g. Inclination (60 degrees)

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TABLE I. Dimensions and weights.

Unit	Width cm 1/ (in.) 2/	Depth 3/ cm (in.)	Height cm (in.)	Weight kg 4/ (lbs) 5/
Receiver	44.13 (17.375)	47.0 (18.5)	17.5 (7)	34.0 (13.6)
Shock and vibration mounting assembly	50.2 (19.75)	43.2 (17)	10.8 (4.25) 6/	7.25 (16)

1/ Centimeters
2/ Inches
3/ Exclusive of rear connectors
4/ Kilograms
5/ Pounds
6/ Maximum unloaded height of bedplate

3.12 Orientation. The equipment shall be capable of storage in any orientation.

3.13 Quantitative reliability requirements. The upper test mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) shall be 2000 hours. The lower test MTBF (θ_1 as defined by MIL-STD-781) shall be 1000 hours.

3.14 Basic conditions. The basic conditions specified in 3.14.1 through 3.14.1.9 shall apply to the performance requirements specified in 3.16, except as modified by a particular test or test procedure specified in this specification. These conditions are specifically applicable to the measurement of spurious response and oscillator radiation characteristics in the equipment.

3.14.1 Standard test conditions. Standard test conditions shall be as specified in 3.14.1.1 through 3.14.1.9.

3.14.1.1 Primary power. The primary power test condition shall be 115 VAC, 60 Hz, single-phase.

3.14.1.2 Ambient temperature. The ambient temperature shall be between 20° Celsius (C) and 30°C.

3.14.1.3 Relative humidity. The relative humidity shall be less than 95 percent.

3.14.1.4 External and internal frequency standard switch. The FREQ STD switch shall be adjusted to the internal or compare position.

3.14.1.5 Internal frequency standard. The internal frequency standard adjustment control shall be adjusted as required to ensure correct setting of the output frequency.

3.14.1.6 Audio frequency (AF) outputs. Unless otherwise specified herein, all AF outputs shall be measured with a 600-ohm nonreactive load connected across the output terminals.

3.14.1.7 RF input to equipment. All RF inputs to the equipment shall be applied to the antenna input terminals through a standard dummy antenna (see 6.7.1).

3.14.1.8 Equipment phone level controls. The equipment phone level controls shall be adjusted as required herein for the determination of performance of the equipment.

3.14.1.9 Equipment beat frequency oscillator (BFO) control. Normally, the BFO control shall be adjusted at the zero or mid-scale position, but may otherwise be adjusted, as specified in 3.16, to determine the performance of the equipment.

3.15 Mechanical design and construction. Mechanical design and construction shall conform to the Mechanical design and construction paragraph of MIL-E-16400, to the extent of being compatible with the GFE.

3.15.1 General requirements. General requirements shall be as specified in 3.15.1.1 through 3.15.1.37.

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3.15.1.1 Construction. Construction shall conform to the Construction paragraph of MIL-E-16400. The equipment shall consist basically of a chassis, a front panel, a case, and a group of replaceable plug-in assemblies. The chassis shall include the power supply, receptacles for insertion of the plug-in assemblies, RF circuit protection, and the mechanical hardware associated with frequency selection. The front panel shall contain the required control functions and shall be securely mounted to the chassis. Two handles shall be provided, one mounted on each side of the front panel and each capable of supporting the entire weight of the unit. All sharp corners and edges on cases, chassis, panels, and brackets shall be eliminated.

3.15.1.2 Case. The case shall be made of aluminum, and shall be of molded construction. A retractable cable and a chassis mating receptacle shall be provided to enable operation of the equipment when the chassis is withdrawn from the case and oriented ±90 degrees from the normal horizontal position. Additional accessory cables shall not be required for operation when the equipment is withdrawn from the case. The retractable cable shall be secured to the chassis by a cable clamp and installed so that the cable has no abrasion points and does not jam when the chassis is inserted in, or withdrawn from, the case.

3.15.1.2.1 Rack mounting. The equipment shall be designed for installation in a standard rack manufactured in accordance with EIA RS-310-C-77. This type of mounting is limited to shore station application. The case of the equipment shall be designed and constructed so that when mounting brackets are installed on the sides of the case, and the equipment is installed in the rack and not subject to shock or vibration, no other support or reinforcing members will be required to preclude deformation of any part of the equipment.

3.15.1.2.2 Cabinet mounting. When the equipment is installed in a CY-4516( )/U Electronic Equipment Cabinet in accordance with MIL-C-24056, Installation Kit, MK-979/URR shall be used. Information is contained in DWG RE-B2683892 and DWG RE-5034104.

3.15.1.3 Enclosure. The enclosure for the equipment shall be in accordance with the Enclosures paragraph of MIL-E-16400. The degree of enclosure of the equipment shall be dripproof (45 degrees).

3.15.1.3.1 External connections. External connections shall be the same as in the GFE (see 3.4b).

3.15.1.4 Bonding and grounding. The equipment shall conform to the Bonding and grounding paragraph of MIL-E-16400, except as specified in 3.15.1.4.1 and 3.15.1.4.2.

3.15.1.4.1 Case bonding. Case bonding and grounding shall be accomplished by preparing mounting surfaces to provide metal contact between the case and mounting bracket, and between the mounting bracket and shock mount. The area around each threaded insert of the case used to secure the mounting bracket to the case shall be free of paint and chemically treated to preserve the metal and provide a ground. Mounting brackets and the shock mount used with the equipment shall be similarly treated to ensure bonding and and grounding.

3.15.1.4.2 Grounding stud. The stud shall be a pressed-in captive insert made integral with the case and constructed of austenitic corrosion-resistant steel. The captive insert shall be flush with the outside surface of the case and shall contain a blind threaded hole to prevent the entrance of moisture into the case. A 25.4 millimeter (mm) (1 in.) diameter area around the threaded insert shall be free of paint and chemically treated to preserve the metal and provide a ground.

3.15.1.5 Corrosion protection and corrosion-resisting treatments. Corrosion protection and corrosion-resisting treatments shall be in accordance with the Corrosion protection/corrosion-resisting treatments paragraph of MIL-E-16400.

3.15.1.5.1 Painting. Painting shall be in accordance with the Painting paragraph of MIL-E-16400, except that the interior surfaces of each metal equipment enclosure shall be painted a dull black to improve thermal efficiency.

3.15.1.6 Frequency selection system. A digital frequency selection system shall be provided. Manually operated knobs on the front panel shall effectively control the frequency and the display of frequency to which the equipment is tuned. Separate controls shall be provided for the 10-MHz, 1-MHz, 100-kHz, 10-kHz, 1-kHz, and 100-Hz frequency selection, and each control shall cover its range in less than one rotation of the knob.

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3.15.1.6.1 Backlash. The backlash in the mechanical tuning arrangement shall not abrogate the requirements for frequency accuracy and stability. End stops shall not be provided, except for the 1-MHz, 10-MHz, and 100-Hz controls. Each digit selected shall include mechanical detent provisions.

3.15.1.6.2 Knob sequence. The sequence of positioning the frequency selector knobs or the manner or speed in which the knobs are rotated shall not affect the performance requirements specified in 3.16. Positioning of the frequency selector knobs more than once shall not be necessary in obtaining the desired frequency.

3.15.1.7 Fastener hardware. Fastener hardware shall conform to the Fastener hardware paragraph of MIL-E-16400, except as otherwise specified herein. All screws, nuts, bolts, and other fastener hardware shall be of austenitic corrosion-resistant steel, except in those applications requiring material capable of withstanding high tensile stresses. Captive-type bolts shall be used on equipment front panels and electronic plug-in assemblies. Cadmium plating is not acceptable on the captive-type bolts used on the equipment front panels and electronic plug-in assemblies.

3.15.1.8 Controls, indicators, and panel layouts. Controls, indicators, and panel layouts, including lights, receptacles, switches, meters (including functions) and fuseholders, shall be the same as furnished in the GFE, unless otherwise specified herein. A dimmer control for the equipment panel lights is not required.

3.15.1.8.1 Mode selector switch. The mode selector switch shall function in a manner which requires a deliberate action in addition to rotation to accomplish power turnoff, which would result in shutdown of the frequency standard. Design of this control shall be included as part of the panel layout, if required. The mode selector switch assembly shall be so designed to permit replacement as a plug-in assembly.

3.15.1.9 Electrical and RF connectors. Electrical and RF connectors shall be provided and shall conform to the Connectors, electrical and Connectors, radio frequency paragraphs of MIL-E-16400. All non-RF connectors shall be of the solder-lug type; crimped pin types shall not be used. Mating connectors for non-RF external case connections shall be provided with metal cable clamps which accommodate standard shipboard cables. All multipurpose connectors shall have 10 percent spare pins to the extent of compatibility with the GFE.

3.15.1.9.1 RF input and output (I/O) connectors. The coaxial RF connectors for the external frequency input and the antenna input shall be Type N. The coaxial RF connector for the internal frequency standard output shall be Type BNC. The receptacles shall be mounted at the rear of the equipment case and shall be suitably labeled. The center pin of the Type N connectors shall be made captive in all directions.

3.15.1.9.2 Assembly (module) connectors. RF inserts, if used, shall be constrained from nonperpendicular movement to ensure proper mating.

3.15.1.10 Time meter. A time meter is not required.

3.15.1.11 Locking devices. Locking devices on external operating controls are not required unless otherwise specified herein.

3.15.1.12 Panel lights. Panel lights shall be easily replaceable from the front panel without requiring removal of the frame from its case.

3.15.1.13 Motors. Motors, used in the equipment, shall conform to the requirements of MIL-E-16400, to the extent of compatibility with the GFE.

3.15.1.14 Fuses. Fuses shall be in accordance with the Fuses and fuseholders paragraph of MIL-E-16400, except that extra fuses are not required. Fuses shall be mounted on the front panel in blown-fuse indicating fuseholders which conform to MIL-F-19207. Spare fuseholders are not required.

3.15.1.15 Filters. Filters used in the sideband selection process shall be electromechanical or crystal filter type.

3.15.1.16 Phone jacks. Phone jacks shall be provided on the front panel to enable the use of earphones in place of a speaker for full operation of the equipment.

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3.15.1.17 Telephone plug. Telephone plugs are not required to be furnished by the contractor.

3.15.1.18 Assembly identification. All assemblies to which the interchangeability requirements apply (see 3.3) shall be provided with the appropriate assigned National Stock Number, Manufacturer's Part Number (including revision indicator), Federal Supply Code Number, and Serial Number. This information shall be inscribed on the baseplate of the assembly.

3.15.1.19 Dust covers. The dust covers of all assemblies employing PCBs shall indicate the location of all test points and all adjustments accessible through the dust cover.

3.15.1.20 Accessibility. In addition to the requirements of the Accessibility paragraph of MIL-E-16400, the circuit mounting boards, components, and tubes shall be easily accessible for maintenance, repair, or replacement. The requirement shall be satisfied to the extent of being compatible with the GFE. Components which require adjustment prior to start of operation shall be accessible without removal of dust covers. All new printed wiring boards shall be removable without the unsoldering of any leads, except 1A1A1A1 MOV PCB and 1A1A11 vernier PCB.

3.15.1.21 Test points. Test points shall be provided and located to permit the isolation of malfunctioning parts or subassemblies within major units or assemblies, without the unsoldering of any interconnection or wiring lead. Test points shall be identical with those in the GFE for existing circuitry.

3.15.1.22 Circuit alignment. Resonant circuits shall be provided with adjustable trimmers, or equally effective means, to permit accurate alignment. The trimmers shall provide sufficient means of adjustment to compensate for the effects of normal aging of electron tubes or other parts.

3.15.1.22.1 Bandpass filters. Composite bandpass filters may be furnished as sealed units without trimmers, provided the I/O sections do not require adjustment.

3.15.1.23 Factory adjustment controls. The design of the equipment shall not include factory or sealed adjustment controls unless specifically approved by the procuring activity.

3.15.1.24 Shock, vibration, and inclination. Shock, vibration, and inclination requirements shall be as specified in 3.15.1.24.1 through 3.15.1.24.5.

3.15.1.24.1 Shock. The equipment shall be capable of withstanding the shock test for Grade A, deck-mounted, Class II, lightweight, Type A equipment as specified in MIL-S-901.

3.15.1.24.2 Vibration. The equipment shall be capable of withstanding the Type I vibration test of MIL-STD-167-1. During vibration testing, the equipment shall conform to the requirements of 3.16.8.1. The identification plates requirements of MIL-STD-167-1 are not applicable.

3.15.1.24.3 Inclination. The equipment shall conform to the inclination requirements specified in 3.11 and the Inclination paragraph of MIL-E-16400.

3.15.1.24.4 Shock and vibration mount distortion. The shock and vibration mounting assembly shall show no deformation, bottoming, or leaning on completion of the shock vibration, or inclination tests.

3.15.1.24.5 Shock and vibration mounting assembly. Shock, vibration, and inclination tests shall be performed, using the shock and vibration mounting assembly specified in 3.8a (see 4.5.6.17).

3.15.1.25 Printed wiring. The fabrication of printed wiring boards and the insertion of parts and assemblies thereon shall conform to the Printed wiring paragraph of MIL-E-16400, to the extent of compatibility with the GFE.

3.15.1.26 Welding. Welding shall conform to the Welding, structural paragraph of MIL-E-16400.

3.15.1.27 Attachment of wires and leads. Attachment of wires and leads shall conform to the Attachment of wires and leads paragraph of MIL-E-16400.

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ration period. The equipment shall be mechanically cycled per... thru the various phases

3.15.1.28 Internal wiring and cabling. Internal wiring and cabling shall conform to the Internal wiring and cabling paragraph of MIL-E-16400. All wiring shall be tetrafluoroethylene insulated. Hookup wire shall conform to MIL-W-16878/4.

3.15.1.29 Conductors. Conductors shall be in accordance with the requirements of MIL-E-16400, except that the number and purpose of conductors shall be the same as those used in the GFE.

3.15.1.29.1 Conductor identification. Identification of conductors shall conform to the Conductor identification paragraph of MIL-E-16400. The method used for identification shall be consistent throughout the equipment.

3.15.1.30 Identification and marking. Identification and marking of all units, assemblies, electrical, and moving mechanical parts, and mechanical parts which require removal for normal maintenance, shall be in accordance with the Identification and marking paragraph of MIL-E-16400. Hertz or Hz shall be used for all indications of frequency on hardware and associated software. Reference designators shall be the same as used in the GFE to the maximum extent practicable.

3.15.1.31 Part replacement. Part replacement shall conform to the Part replacement paragraph of MIL-E-16400.

3.15.1.32 Quartz crystal units. Quartz crystal units shall conform to the Quartz crystal units paragraph of MIL-E-16400.

3.15.1.33 Gears and cams. Gears and cams shall conform to the Gears and cams paragraph of MIL-E-16400, except that only metal gears shall be used.

3.15.1.34 IF and audio assembly adjustment. Adjustable controls on the IF and audio assemblies which are used to set IF gain and automatic gain control on the IF and audio assemblies which are used to set IF gain and automatic gain control (AGC) thresholds shall be accessible from the top without removal of the dust cover.

3.15.1.35 Maintenance accessories. Extender cables or extender cards, as applicable, shall be furnished as required (see 6.2.1). Electronic assemblies and PCBs that plug in to chassis or mother board receptacles and on which the test points or components are not available for fault isolation, require an extender cable or extender card for fault isolations.

3.15.1.35.1 Extender cables. Extender cables for the assemblies listed in 3.4.1a, and for new assemblies, shall be 610 mm (24 in.) long.

3.15.1.35.2 PCB extender cards. The extender cards shall be of sufficient height to permit access to all test points and components on the PCB for dynamic fault isolation. The extender cards shall conform to the Printed wiring paragraph of MIL-E-16400.

3.15.1.36 Frequency standard assembly construction (maintenance considerations). Construction of the frequency standard assembly shall permit maintenance or replacement, or both, of: a) the oscillator and oven control circuits and, b) the oven case as separate plug-in subassemblies. The metal oven case subassembly components (except the heater winding) shall be readily removable for replacement. No potting or molding shall be employed to secure replaceable components to the oven case. Nylon screws shall not be used.

3.15.1.37 Thermal design. Thermal design shall be in accordance with the Thermal design paragraph of MIL-E-16400 to the extent of compatibility with the GFE.

3.15.2 Human engineering. The contractor shall comply with the human engineering operational and maintenance design criteria of MIL-STD-1472 and the requirements specified herein, except that equipment shall be designed for use by the 5th percentile female to the 95th percentile male.

3.15.3 Shock and vibration mounting assembly. The shock and vibration mounting assembly shall conform to the requirements of MIL-E-16400. The mount, when attached to the equipment, shall enable the equipment to conform to the requirements of 3.15.1.24.

3.16 Performance. Performance requirements shall be as specified in 3.16.1 through 3.16.17.

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3.16.1 Sensitivity and gain. The sensitivity and gain requirements of 3.16.1.1 through 3.16.1.3 shall apply to the equipment.

3.16.1.1 Sensitivity. The sensitivity of the equipment (see 6.7.7) shall conform to the requirements specified in TABLE II for any frequency within the complete frequency range of the equipment. Specified sensitivities shall be achieved in all modes with the manual RF gain control in the fully clockwise (maximum gain) position.

TABLE II. Sensitivity.

Mode	CW	MCW, AM	USB, LSB, ISB, RATT
IF bandwidth (kHz)	6.0	6.0	3.0
Sensitivity μV 1/	1.8	6.0	1.2

1/ Microvolts

3.16.1.2 Gain. The equipment gain shall be sufficient to produce standard audio output (see 6.7.3) for the input voltages of TABLE II, for any frequency within the complete frequency range of the equipment.

3.16.1.3 Gain variation. The variation in overall gain for any mode of reception shall not exceed 10 decibles (dB) for the complete working frequency range of each MHz frequency band. Gain variation over the complete equipment tuning range (2 MHz to 30 MHz) shall not exceed 15 dB total.

3.16.2 Selectivity (see 6.7.8). Selectivity requirements shall be as specified in 3.16.2.1 through 3.16.2.5.

3.16.2.1 Selectivity (antenna input to detector input) (see 6.7.8.1). The selectivity of the equipment, excluding the AF response characteristics, shall depend primarily upon the characteristics of the filter defining the final IF passband. When measured in accordance with 4.5.6.2, selectivity shall be as specified in 3.16.2.1.1 and 3.16.2.1.2.

3.16.2.1.1 USB, LSB, and ISB. USB, LSB, and ISB passband characteristics shall be as specified in a through h:

a. USB passband (-3 dB): Carrier +300 Hz to carrier +3500 Hz
b. LSB passband (-3 dB): Carrier -300 Hz to carrier -3500 Hz
c. ISB passband (-3 dB): Carrier +300 Hz to carrier +3500 Hz, and carrier -300 Hz to carrier -3500 Hz
d. Ripple: 3 dB peak-to-valley, maximum
e. Attenuation at carrier frequency: 15 dB minimum
f. Opposite sideband attenuation 1 kHz into opposite sideband: 60 dB minimum
g. Attenuation at carrier +5300 Hz (USB): 60 dB minimum
h. Attenuation at carrier -5300 Hz (LSB): 60 dB minimum

3.16.2.1.2 AM, CW. AM, CW shall be as specified in a through c:

a. Passband (-3 dB): Carrier -3500 Hz to carrier +3500 Hz
b. Ripple (peak-to-valley): 3 dB maximum
c. Attenuation at carrier +9.5 kHz, carrier -9.5 kHz: 60 dB minimum

3.16.2.2 Overall passband (antenna input to audio output) (see 6.7.8.2). When measured in accordance with 4.5.6.2.1, the overall equipment passband characteristics shall be as specified in 3.16.2.2.1.

3.16.2.2.1 USB, LSB, and ISB. USB, LSB, and ISB equipment passband characteristics shall be as specified in a through d:

a. USB passband: Carrier +300 Hz to carrier +3500 Hz
b. LSB passband: Carrier -300 Hz to carrier -3500 Hz
c. ISB passband: Carrier +300 Hz to carrier +3500 Hz, and carrier -300 Hz to carrier -3500 Hz

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d. Ripple (peak-to-valley): Ripple (peak-to-valley) shall be as specified in 1 through 4:

1. Carrier +300 Hz to carrier +500 Hz: 5 dB maximum USB
2. Carrier +500 Hz to carrier +3500 Hz: 3 dB maximum USB
3. Carrier -300 Hz to carrier -500 Hz: 5 dB maximum LSB
4. Carrier -500 Hz to carrier -3500 Hz: 3 dB maximum LSB

3.16.2.3 Overload selectivity (desensitization) (see 6.7.8.3). Overload selectivity shall be such that when the equipment is operated in SSB modes, and with AGC on (slow), an unmodulated out-of-band interfering signal at the specified frequencies and amplitudes shall not cause a decrease of more than 3 dB in the signal plus-noise-to-noise (S+N/N) ratio due to an in-band signal at the level of standard sensitivity specified in a and b:

a. $\frac{f_o}{1.05} \geq f_s \geq 1.05 f_o$: $P_S \leq$ -1 decibel referred to one milliwatt dBm

b. $\frac{f_o}{1.12} \geq f_s \geq 1.12 f_o$: $P_S \leq +29$ dBm

Where f_o is the operating frequency, f_s is the out-of-band signal frequency, and P_S is the out-of-band available power.

3.16.2.4 Gain depression. At the desensitization points specified in 3.16.2.3, gain depression shall not reduce the desired output by more than 8 dB.

3.16.2.5 Envelope delay distortion. The maximum envelope delay distortion shall not exceed 500 microseconds (µs) throughout the region from 825 Hz to 3100 Hz nor exceed 150 µs for any 100 Hz interval within this range. Maximum envelope delay distortion shall not exceed 700 ms in the region from 605 Hz to 3100 Hz, or 1.5 milliseconds (ms) over the complete range from 380 Hz to 3100 Hz.

3.16.3 Spurious responses. The equipment shall conform to the requirements of 3.16.3.1 through 3.16.3.6. The definitions of 6.7 are applicable.

3.16.3.1 Internally generated spurious outputs. With no RF input signal, the antenna input terminated with a shielded 50-ohm resistor, and the equipment adjusted for the reception of SSB signal, internally generated equipment outputs shall conform to the requirements of 3.16.3.1.1 and 3.16.3.1.2.

3.16.3.1.1 Equipment operated on an internal frequency standard. No more than five spurious responses shall occur with output levels equivalent to that resulting from a 0.4 µV to 4.0 µV (open circuit) reference signal applied to the antenna input. All other spurious responses generated with the equipment shall have output levels less than that resulting from a 0.4 µV (open circuit) reference signal applied to the antenna input.

3.16.3.1.2 Equipment operated on an externally supplied 5 MHz reference. When the equipment is supplied an external 5 MHz reference at an available power of +21 dBm (5 V open circuit in series with 50 ohms), having all nonharmonically related components at least 90 dB, and all harmonically related components at least 60 dB, below the reference, internally generated spurious outputs shall conform to the requirements of 3.16.3.1.1, except that six spurious responses shall be permitted with output levels equivalent to a 0.4 µV to 4.0 µV (open circuit) reference signal applied to the antenna input.

3.16.3.2 Image rejection. Rejection of the first mixer image shall be not less than 100 dB for any equipment resonant frequency below 20 MHz. First mixer image rejection shall be not less than 90 dB for equipment resonant frequencies of 20 MHz and above. Rejection of second and third mixer images shall be not less than 100 dB.

3.16.3.3 IF rejection. Rejection of the first IF shall be not less than 100 dB for any equipment resonant frequency below 22 MHz. First IF rejection shall be not less than 80 dB for resonant frequencies of 22 MHz and above. For any operating frequency which differs from the corresponding second IF by more than 5 percent, second IF rejection shall be not less than 100 dB. Third IF rejection shall be not less than 100 dB of all operating frequencies.

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3.16.3.4 Other single-signal spurious responses. Single-signal spurious responses, other than IF and image, shall be attenuated in accordance with the requirements specified in 3.16.3.4.1 through 3.16.3.4.3.

3.16.3.4.1 Operating frequency below 15 MHz. Operating frequency below 15 MHz shall be as specified in a through c:

a. $\frac{f_o}{1.025} < f_s < 1.025f_o$: 60 dB

b. $\frac{f_o}{1.025} \geq f_s \geq 1.025f_o$: 100 dB

c. $\frac{f_o}{1.10} \geq f_s \geq 1.10f_o$: 140 dB

3.16.3.4.2 Operating frequency 15 MHz to 20 MHz. Operating frequency 15 MHz to 20 MHz shall be as specified in a through c:

a. $\frac{f_o}{1.025} \leq f_s < 1.025f_o$: 60 dB

b. $\frac{f_o}{1.025} \geq f_s \geq 1.025f_o$: 90 dB

c. $\frac{f_o}{1.12} \geq f_s \geq 1.12f_o$: 140 dB

3.16.3.4.3 Operating frequency greater than 20 MHz. Operating frequency greater than 20 MHz shall be as specified in a through c:

a. $\frac{f_o}{1.025} < f_s < 1.025f_o$: 60 dB

b. $\frac{f_o}{1.025} \geq f_s \geq 1.025f_o$: 80 dB

c. $\frac{f_o}{1.15} \geq f_s \geq 1.12f_o$: 140 dB

3.16.3.5 Front-end intermodulation (out-of-band). Signals at two frequencies, f1 and f2, shall be applied simultaneously to the antenna input from a 50-ohm linear combiner. Sufficient filtering or isolation shall be used between each generator and the combiner to ensure adequate freedom from intermodulation products in the test signal. Signal pairs shall be $(f_o + \Delta f, f_o + 2\Delta f)$, $(f_o - \Delta f, f_o - 2\Delta f)$, $(f_o + 2\Delta f, f_o + 3\Delta f)$, and $(f_o - 2\Delta f, f_o - 3\Delta f)$, with Δf equal to 5 percent f_o. A reference signal of 1.2 µV (open circuit) shall be applied at a frequence which results in a 1-kHz audio output, and the reference output amplitude recorded. The reference signal shall then be replaced by the signal pairs which shall be increased in amplitude, maintaining equality of the two signals, until the equipment output reaches the reference output amplitude, or until each input signal amplitude reaches 0.1 V (open circuit). The equipment audio output shall not reach the reference for input signal amplitudes (each signal) of 0.1 V (open circuit) or less.

3.16.3.6 Cross modulation. Cross modulation selectivity shall be such that an off-resonant signal, 90 percent modulated, with a carrier amplitude of 5.0 V open circuit, or less, and spaced 15 percent or more from the resonant frequency, shall not produce an output greater than 10 dB below that resulting from the resonant signal. For operating frequencies below 25 MHz, this requirement also applies when the frequency separation is 2.5 percent, or more, and the interfering signal carrier level is 0.1 V open circuit, or less. For operating frequencies above 26 MHz, the requirement also applies for a frequency separation of 4.0 percent, or more, and an interfering signal carrier level of 0.1 V, or less. AGC shall be on (slow).

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3.16.4 Audio outputs. The audio outputs from the USB and LSB amplifiers shall conform to the requirements of 3.16.4.1 through 3.16.4.4.2. Each amplifier shall feed line output receptacles at the rear of the equipment and the own head telephone jack on the front panel. The line outputs shall terminate in appropriately identified receptacles. Transformer couplings shall be provided between each AF amplifier and associated head telephone jack and AF line output connector. The secondary windings feeding power to the output connector and the head telephone jack shall be electrostatically shielded from their associated primary windings. Secondary windings which feed AF power to the rear AF line output connectors shall be electromagnetically balanced with respect to ground. The center tap shall be brought out to a separate terminal on the equipment chassis.

3.16.4.1 Audio output voltage regulation. Each audio amplifier shall be so designed that the output voltage shall remain constant within ±30 percent (in V) of the value obtained with a 600-ohm nonreactive load for a change in the resistive value of this load from 100 ohms to 600 ohms in 100-ohm steps. This condition shall be obtained with any input level from zero input to that which produces full output. The power output transistors shall not be subjected to damage or deterioration at any output load impedance from zero ohms to an open circuit.

3.16.4.2 Maximum undistorted output. With an RF input of not less than 1000 µV, the maximum undistorted output (see 6.7.11) shall be not less than 60 milliwatts (mW), dissipated in a 600-ohm resistive load at the remote output receptacles of the audio amplifier; or with the line level control set at the maximum output settings, the undistorted output appearing across a 600-ohm resistive load connected to the phone jack shall be not less than 15 mW. In the AM mode, with 75 percent modulation, distortion shall be not greater than 5 percent.

3.16.4.3 Audio level controls. Separate (noncoaxial) level controls shall be provided on the equipment front panel for audio line and local (phone) outputs. Phone level controls shall be equipped with knobs chosen in accordance with MIL-E-16400 part selection procedures. Line level controls shall be screwdriver-adjustable types with shaft locks. All controls shall be in accordance with the requirements of MIL-R-94. Controls shall be standard audio (clockwise log) taper, and shall provide smooth variation in output over a minimum range of 40 dB.

3.16.4.3.1 Audio level metering. The equipment shall be provided with a front panel audio metering system which will indicate audio line output levels. A single meter shall be used with suitable switching to permit indication of either sideband output to within +1 dB over an amplitude range of -20 dBm to +22 dBm, and a frequency range of 300 Hz to 3500 Hz. The meter shall be in accordance with MIL-M-10304, and the required amplitude range, resolution, and accuracy shall be provided by a meter with a single scale with range switching in 10 dB steps.

3.16.4.4 Output hum and noise. Output hum and noise shall be as specified in 3.16.4.4.1 and 3.16.4.4.2.

3.16.4.4.1 Output signal-to-hum ratio. The output signal-to-hum ratio for AM, USB, or LSB shall be not less than 45 dB for the standard audio output of 6 mW. This requirement shall apply separately to each output component which is related to the power line frequency, including hum sidebands on the desired audio output. Measurement shall be in accordance with 4.5.6.4.5.

3.16.4.4.2 Output signal-to-hum-plus-noise ratio. The output signal-to-hum-plus-noise ratio for AM, USB, and LSB modes, measured in accordance with 4.5.6.4.5, shall be not less than 35 dB.

3.16.5 Manual RF gain control. The manual RF gain control shall have the range to provide a variation in the overall gain of the RF and IF amplifiers of the equipment of not less than 125 dB. The 125 dB of attenuation shall be obtained in not less than 150 degrees of angular rotation. Excluding the first 10 dB of gain reduction at the clockwise limit of the control, and the last 10 dB of gain reduction at the counterclockwise end of the control, the maximum rate of change of gain shall be 1.2 dB per degree of control rotation. The manual RF gain control shall be inoperative when the equipment AGC switch is in SLOW and FAST.

3.16.6 AGC characteristics. The AGC threshold shall be adjustable over a minimum range of 1.0 µV to 5.0 µV, open circuit. Production adjustment shall result in the initial characteristics specified in a through c:

a. The increase in audio output shall not exceed 6 dB for an RF input change from 1.2 µV to 5.0 µV, open circuit

b. Audio output shall remain constant within a total tolerance of 3 dB for RF input variations from 5.0 µV to 100,000 µV, open circuit

c. The total harmonic distortion shall not increase above that in 6.7.11 at RF input levels up to 100,000 µV, open circuit

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3.16.6.1 AGC conditions. The requirements of 3.16.6 shall apply for the conditions specified in a through d, singly or in combination:

a. Any frequency selected within the nominal frequency range of the equipment
b. Any audio gain adjustment producing audio levels at or below maximum undistorted output
c. Any constant level of carrier modulation between 30 percent and 90 percent
d. Any constant modulation frequency between the limits of 300 Hz and 3500 Hz

3.16.6.2 AGC time constants. The AGC time constants shall be as specified in TABLE III. These time constants shall hold for signal transitions of 60 dB or less.

TABLE III. AGC time constants.

AGC switch position	Attack time (ms)	Hang time (ms)	Decay time (ms)
Fast (RATT)	-	NLT 1/ 10	NMT 2/ 15
Data (NTDS)	NMT 15	NMT 25	
Slow (SSB VOICE, CW, AM)	NMT 20	NMT 800	
		NMT 1200	NMT 400
OFF	-	-	-

1/ Not less than
2/ Not more than

3.16.6.2.1 Independence of AGC switch position. The AGC switch positions shall be completely independent of the mode selector switch positions. All time constants (see TABLE III) shall be switchable to all mode selector switch positions.

3.16.6.2.2 Location of AGC switch. The vertical location of the AGC switch on the front panel shall be such as to allow for the future addition of a similar control.

3.16.6.3 AGC linearity. There shall be no evidence of blocking or amplifier saturation at any input level from standard sensitivity up to 1 V.

3.16.6.4 Resonant overload. Audio output power (remote outputs) at resonant overload (see 6.7.10) shall be a minimum of 60 mW.

3.16.6.5 Time constant of recovery from surge inputs. The design shall be such that with the equipment operating under standard conditions for the reception of A1 signals, and with AGC off, a sharp pulse of CW input signal, at resonance, having a minimum amplitude of 5 V and duration of 50 ms (to simulate static surge), will not cause a reduction in sensitivity that will persist in excess of 30 ms after cessation of the input surge.

3.16.6.5.1 Measurement of time constant. The time constant shall be measured from the time of removal of the input signal pulse until the output noise level has reached 63 percent of the steady-state value. Start of time shall begin at the moment of removal of the input signal pulse.

3.16.7 BFO control. The BFO control shall operate (when the mode switch is set at CW) so as to permit a variation of the frequency of the BFO of not less than ±3000 Hz.

3.16.7.1 BFO frequency variation. The variation of frequency versus angular rotation of the BFO shall be substantially linear. The frequency of the BFO shall increase in the clockwise direction and decrease in the counterclockwise direction of the control, using the center of rotation as zero reference.

3.16.7.2 BFO buffering. The BFO shall be buffered to prevent pulling of the BFO frequency by more than ±10 Hz due to equipment RF input signal amplitudes of 10,000 µV or less.

3.16.8 In-band intermodulation. Intermodulation distortion products generated when the equipment processes two equal-amplitude modulated RF signals producing audio outputs at 935 Hz and 1045 Hz shall each be a minimum of 33 dB below the reference output tones. This requirement shall be satisfied in USB and LSB operation with the AGC time constant selector switch in both FAST and SLOW positions, over an input amplitude range of 100 µV to 50,000 µV open circuit, per tone, and for audio output levels of +10 dBm or less, per tone.

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3.16.8.1 Vibration test conditions. During vibration tests, the equipment shall be operated in the USB mode with the AGC time constant selector in the SLOW position, and with a single phase-stable unmodulated RF input signal producing an audio output of 1000 Hz to 3000 Hz. The RF input signal amplitude shall be 1000 µV, open circuit, and the resulting audio output signal shall be continuously monitored with an oscilloscope, wave analyzer, and a phase detector. Any vibration-related product present at the audio output shall be not less than 30 dB below the reference audio output amplitude. No signal loss (dropout) exceeding 3 dB may occur during the vibration test. Phase stability shall satisfy the requirements of 3.16.11.1.1.

3.16.9 RF input circuit. The equipment antenna input circuits shall be designed for maximum sensitivity to signals from a 50-ohm impedance. The input circuits shall be protected from RF voltages up to 100 volts root-mean-square (Vrms) at the antenna input connector of the equipment. The protective circuit operating threshold shall be between 6.0 Vrms and 7.0 Vrms, closed circuit, measured with a high impedance RF voltmeter connected directly to the protection circuit input terminal and with output terminal terminated in a 50-ohm resistive load. The antenna overload circuit shall also be operated by changes in the MHz tuning selectors, providing the designed decoupling during rotation of the MHz selection motors in the equipment. The overload protection circuit shall automatically reset when the overload voltage no longer exists. At least the above degree of protection shall exist with the equipment on or off. Fuses are not acceptable for RF overload protection. Operation of the protection circuit by an out-of-band transmitter shall not produce more than 40 dB desensitization at any equipment operating frequency; measurement shall be as specified in 4.5.6.9. Equipment sensitivity shall not degrade by more than 1 dB relative to reference measurements made prior to testing in accordance with 4.5.6.9.

3.16.10 Accuracy of main tuning indicator calibration. The accuracy of frequency indication shall be within the limits specified in 3.16.10.1 and 3.16.10.2. The specified accuracies shall apply to all tuning indicators acting in combination, to indicate the tuning of the equipment.

3.16.10.1 Tuning steps. Tuning in steps of 100 Hz shall be provided with an accuracy equivalent to that of the internal frequency standard. Tuning the equipment to any frequency within the tuning range in less than 20 seconds shall be possible.

3.16.10.2 Frequency vernier. A control shall be provided to permit switching into a vernier tuning mode to provide continuous vernier tuning between 1 kHz points. Frequency stability shall be such that for a fixed RF input frequency in SSB modes, audio output frequency will not vary more than 5 Hz in 10 minutes of vernier operation, after a stabilization period of 1 hour. A flashing amber warning light shall indicate that the vernier control is in the ON position. The vernier control shall be uncalibrated.

3.16.11 Frequency and gain stability. The equipment shall include provisions for operation from either an internal or external frequency standard of 5 MHz. All frequency stability requirements apply to operation using the internal standard.

3.16.11.1 Frequency stability and gain variations. The requirements of 3.16.11.1.1 through 3.16.11.1.6.2 for frequency stability and gain variation shall apply under conditions of A1 or A9 reception, as specified. The frequency stability figures shall apply to the variation in a 1000-Hz AF beat note with a constant input frequency. The frequency changes shall be with respect to the effects or results of the frequency variation of all oscillators on the audio beat note frequency. All figures shall refer to total variations, either instantaneous, transient, or gradual changes. The requirements specified herein shall apply at any frequency within the frequency range of the equipment.

3.16.11.1.1 Sixty-minute warm-up period. Measurements shall be made in the A9 mode at constant ambient temperature, constant relative humidity, and constant line input voltage. Two minutes after turn-on time, the equipment shall be within 0.1 percent of the frequency reached at the end of 60 minutes. Fifteen minutes after turn-on time, the equipment shall be within 0.002 percent of the frequency reached 60 minutes after turn-on. Sixty minutes after turn-on, the equipment shall be within 1 part in 10^7 of the desired frequency.

3.16.11.1.2 Variation in primary power voltage and frequency. Variation of primary power supply line voltage over a range of 115 V, ±10 percent, and of power line frequency over a range of 48 Hz to 420 Hz, in any combination, shall not cause a change in resonant frequency of the equipment by more than 2 parts in 10^8 (total) for A9 signals, or a change in overall gain (AGC defeated) in excess of 2 dB.

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3.16.11.1.3 Variation in relative humidity. Exposure of the equipment to the humidity conditions specified in the Humidity paragraph of MIL-E-16400 shall not cause loss of synthesizer frequency lock or excessive changes in frequency or gain. During the checks performed at the end of the second and fifth cycle of the humidity test, the equipment shall be energized for a maximum period of 30 minutes to verify proper operation. At the end of the 30-minute period, frequency for SSB modes shall be within 2 parts in 10^7 of the nominal, and maximum permissible gain change (AGC OFF) shall not cause audio output levels to vary by more than 8 dB (total) from those recorded as references at the end of the initial dry-out period.

3.16.11.1.4 Variations in ambient temperature. Variations in ambient temperature shall be as specified in 3.16.11.1.4.1 and 3.16.11.1.4.2.

3.16.11.1.4.1 Nonoperating condition. The equipment shall be capable of normal operation after exposure to the temperature cycle specified in MIL-E-16400 for temperature tests (Range 4) to be applied to the equipment when in a nonoperating condition.

3.16.11.1.4.2 Operating condition. The equipment, when subjected to the temperature tests for Range 4 equipment specified in MIL-E-16400 (when in the operating condition), shall not exhibit frequency or gain variation in excess of the limits specified in TABLE IV. The specified limits for variation of frequency and gain shall be the total deviations of these parameters, which occur during the time period required by the high and low operating temperature cycling test procedures specified in MIL-E-16400. The AGC shall be off during gain variation measurements.

TABLE IV. Frequency and gain variation.

Performance requirement	A1	A9
Frequency variation versus change in ambient temperature from 0°C to +50°C	400 Hz (total)	1 part in 10^7
Gain variation versus change in ambient temperature from 0°C to 50°C	8 dB (total)	8 dB (total)

3.16.11.1.5 Control variations. Frequency variations resulting from a variation of any nonfrequency determining control throughout the full range of operation shall not exceed ±75 Hz for A1 and 1 part in 10^8 for A9. This requirement shall apply to those controls which operate in nonfrequency determining circuits, and to controls which affect the RF and AF gain or level. The specified limit for frequency variation shall be the absolute maximum deviation of this parameter which occurs during the manipulation of controls throughout the full range of operation.

3.16.11.1.6 Short and long term stability. The equipment shall be capable of operation for a period of 24 hours without operator attention. Without operator attention shall be interpreted as meaning that reliable reception of a desired signal shall be assured throughout a 24-hour period without the necessity for readjustment of any tuning control. This requirement shall be applicable throughout the complete frequency range of the equipment.

3.16.11.1.6.1 Sudden changes during 24-hour period. During the 24-hour period of operation specified in 3.16.11.1.6, the equipment shall not exhibit sudden sporadic shifts in the frequency generated by any of the oscillators, or sudden changes in gain that may be the result of easing tensions in tuning drive mechanisms or from similar effects.

3.16.11.1.6.2 Frequency drift rate. After an initial stabilization period, not to exceed 48 hours following turn-on, the overall frequency drift rate shall not exceed 1 part in 10^8 (total) per day. This rate shall be achieved throughout the equipment operating frequency range, in any mode of operation which does not use the BFO or frequency vernier. The long-term frequency drift rate, averaged over a period of 60 days, shall not exceed 2.5 parts in 10^9 (total) per day. During measurement of frequency drift rate, changes in power supply line voltage and frequency shall be limited to ±1 percent, and temperature variations shall be limited to 10°C (total) under the same conditions, equipment gain (AGC off) shall not change by more than 3 dB (total) during a 24-hour period after initial stabilization.

3.16.11.1.7 Phase stability. Phase stability shall be such that the probability of experiencing spurious phase changes greater than 5 degrees between successive frames of 13.3 ms or 22 ms shall be less than 1 percent. Measurement shall be the average phase in a central 9.09 ms or 18.18 ms period for 13.3 ms or 22 ms frame times, respectively. Data shall be acquired for a sufficient number of adjacent frame pairs to establish the specified probability with a confidence of at least 95 percent.

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3.16.12 Regeneration. There shall be no evidence of unintended oscillations or instability under any condition of equipment operation, including primary line voltage 10 percent higher than the nominal primary line voltage.

3.16.13 Oscillator radiation. Oscillator radiation shall conform to MIL-STD-461, Class A4 requirements. Antenna terminal narrowband emissions in the region 10 kHz to 1 GHz shall not exceed -90 dBm. Broadband emissions in the same frequency range shall conform to the requirements of MIL-STD-461, Method CS06.

3.16.14 Frequency standard input, calibration, and adjustment. The equipment shall contain provisions that permit the comparison of an external higher stability frequency standard input of 5 MHz with the internal frequency standard of the equipment. The comparator shall function with external signal levels between 450 millivolts (mV) and 10.0 V, open circuit, in series with 50 ohms.

3.16.14.1 Frequency standard calibration. The internal frequency standard shall contain a means of indicating calibration accuracy when compared to an external frequency standard. A dial shall be provided to indicate positioning of the frequency control circuit element within the adjustment range.

3.16.14.1.1 Frequency standard adjustment range. The internal 5-MHz frequency standard shall be capable of adjustment over a minimum frequency range of 5 MHz ±20 Hz. The adjustment shall be accessible from the top surface of the frequency standard assembly without removing the assembly cover. Resolution of the adjustment shall be adequate to permit setting within 1 part in 10^9. Piston type adjustments shall be protected from damage at the ends of the adjustment ranges.

3.16.14.2 Operation using an external frequency standard. The equipment shall be capable of operation from an external frequency standard of 5 MHz with any input level between 500 mV and 10.0 V, open circuit, in series with 50 ohms. When the equipment is operated from an external frequency standard and the level of the externally supplied 5-MHz signal falls below 450 mV, open circuit, in series with 50 ohms, the internally contained 5-MHz crystal oscillator shall be automatically selected as the reference (see 3.16.14.4). When the externally supplied 5-MHz signal is restored to a level greater than 800 mV ±100 mV, the external 5-MHz signal shall again be automatically chosen as the reference. Input impedance at the external 5-MHz input terminals shall be 50 ohms, nominal, within a voltage standing wave ratio of 1.5:1 or less.

3.16.14.3 Reference output signal. The equipment shall provide a 5-MHz reference output signal of between +7 dBm and +13 dBm, delivered to a 50-ohm load, for external use.

3.16.14.4 Frequency standard switch functions. The internal frequency standard assembly shall have a function switch located on the top of the assembly, accessible without removal of the assembly dust cover. The switch shall have three positions with the functions (clockwise) specified in a through c:

a. EXT OVEN STBY (external oven standby): Provides operation from an external frequency distribution system; the internal frequency standard oscillator and oven are kept on for immediate availability with full accuracy

b. EXT NORM (external normal): Provides operation from an external frequency distribution system, but the internal frequency standard oscillator and oven are turned off (the normal method of operations in installations having such distribution systems)

c. INT OR COMP (internal or compare): Used when external frequency distribution system is not available or not functioning, or for testing.

3.16.15 Frequency generator. The frequency generator process in the translator synthesizer assembly shall utilize only a single reference frequency, with all internally-generated frequencies used in the synthesis of the operating frequency derived by arithmetic operations on, and phase-locked to the reference. The synthesizer shall not use multi-crystal generators or motor-driven frequency selectors.

3.16.16 Electromagnetic compatibility (EMC). The equipment shall conform to the requirements of MIL-STD-461 for Class A4, conducted emission equipment. Tests CE01 and CE03 shall be performed at both 60 Hz and 400 Hz power line frequencies, and the equipment shall conform to the requirement CS09.

3.16.17 Accelerated life. The equipment shall conform to the Accelerated life paragaraph of MIL-E-16400.

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3.17 Quantitative maintainability requirements. This requirement applies to the GFE design. If the GFE design does not conform to the requirement, redesign of the equipment shall preserve compatibility with the GFE. The equipment shall have a mean-corrective-maintenance-time not exceeding 1.5 hours and a maximum-corrective-maintenance-time (M_{maxct}) not exceeding 4.5 hours when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, PCBs) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The mean-time-to-repair (MTTR) includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions. The equipment is in a corrective maintenance mode whenever a failure, including a malfunction that allows degraded operation, occurs.

3.18 Workmanship. Workmanship shall be as specified in 3.18.1 and 3.18.2.

3.18.1 General workmanship. The workmanship and soldering requirements shall be in accordance with the Workmanship and Soldering paragraphs of MIL-E-16400.

3.18.2 Workmanship screen. All equipment shall withstand a defect detection vibration screen of random type vibration (see 4.5.3.1) at $0.04g^2$/Hz ±3 dB from 80 Hz to 350 Hz and temperature cycling as specified in 4.5.3.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 General inspection. Inspection shall be as specified in MIL-E-16400, except as otherwise specified in 4.2.1 through 4.4.4.

4.2.1 Classification of inspections. The inspection requirements specified herein are classified as specified in a through e:

a. First article inspection (see 4.3)
b. Quality conformance inspection:
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
   3. Environmental inspection (Group C) (see 4.4.3)
c. Reliability testing (see 4.6)
d. Maintainability demonstration (see 4.7)
e. Inspection of preparation for delivery (see 4.8)

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4.3 First article inspection. Unless otherwise specified (see 6.2.1), two units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE V. The inclination test will be performed by the Navy during Government first article testing, and is not required to be performed by the contractor.

TABLE V. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination	3.1.2	4.5.1	X	X		
Weight	3.10	4.5.1	X	X		
Dimensions	3.10	4.5.1	X	X		
Parts, materials, and processes	3.1.2	4.5.1	X	X		
Finish	3.15.1.5	4.5.1	X	X		
Identification and marking	3.15.1.30	4.5.1	X	X		
General workmanship	3.18.1	4.5.2.1	X	X		
Welding	3.15.1.26	4.5.1	X	X		
Mechanical design and construction	3.15	4.5.1	X	X		
Drawer orientation	3.15.1.1	4.5.6.21	X		X	
Bonding and grounding	3.15.1.4	4.5.1	X		X	
Corrosion protection	3.15.1.5	4.5.1	X		X	
Painting	3.15.1.5.1	4.5.1	X		X	
Mode selector switch	3.15.1.8.1	4.5.6.22	X		X	
Assembly identification	3.15.1.18	4.5.6.23	X		X	
Soldering	3.18.1	4.5.1	X		X	
Conductor identification	3.15.1.29.1	4.5.1	X		X	
Frequency standard construction	3.15.1.36	4.5.6.15	X		X	
Frequency range	3.5	4.5.6.10	X	X		
Overload protection	3.6.2	4.5.1	X		X	
Leakage current	3.6.3	4.5.4	X		X	
Sensitivity	3.16.1.1	4.5.6.1	X	X		
Gain	3.16.1.2	4.5.6.1.1	X		X	
Gain variation	3.16.1.3	4.5.6.1.2	X	X		
Selectivity	3.16.2	4.5.6.2	X			
Overall passband	3.16.2.2	4.5.6.2.1	X	X		
Desensitization	3.16.2.3	4.5.6.2.2	X		X	
Gain depression	3.16.2.4	4.5.6.2.3	X		X	
Envelope delay distortion	3.16.2.5	4.5.6.2.4	X		X	
Spurious responses	3.16.3	4.5.6.3	X		X	
Internal spurious	3.16.3.1	4.5.6.3.2	X	X		
Image rejection	3.16.3.2	4.5.6.3	X	X	X	
IF rejection	3.16.3.3	4.5.6.3	X	X	X	
Single-signal responses	3.16.3.4	4.5.6.3	X	X	X	
Front end intermodulation	3.16.3.5	4.5.6.3	X		X	
Cross modulation	3.16.3.6	4.5.6.3.1	X		X	
Audio output regulation	3.16.4.1	4.5.6.4.1	X		X	
Maximum undistorted output	3.16.4.2	4.5.6.4.2	X	X	X	
Audio level controls	3.16.4.3	4.5.6.4.3	X		X	
Audio level metering	3.16.4.3.1	4.5.6.4.4	X		X	
Signal-to-hum ratio	3.16.4.4.1	4.5.6.4.5	X	X	X	

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TABLE V. Examinations and tests. (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Signal-to-hum plus noise ratio	3.16.4.4.2	4.5.6.4.5	X		X	
Manual RF gain control	3.16.5	4.5.6.5	X		X	
AGC characteristics	3.16.6	4.5.6.6	X	X	X	
AGC time constants	3.16.6.2	4.5.6.6.1	X		X	
Resonant overload	3.16.6.4	4.5.6.6.2	X		X	
Surge recovery	3.16.6.5	4.5.6.6.3	X		X	
BFO control	3.16.7	4.5.6.7	X	X		
In-band inter-modulation	3.16.8	4.5.6.8	X		X	
RF input circuit	3.16.9	4.5.6.9	X		X	
Accuracy of tuning	3.16.10	4.5.6.10	X	X		
Tuning time	3.16.10.1	4.5.6.10.1	X		X	
Vernier operation	3.16.10.2	4.5.6.10.2	X		X	
Vernier stability	3.16.10.2	4.5.6.11.7	X		X	
Frequency stability	3.16.11	4.5.6.11	X		X	
Gain stability	3.16.11	4.5.6.11	X		X	
Sixty-minute warm-up	3.16.11.1.1	4.5.6.11.1	X		X	
Line voltage variation	3.16.11.1.2	4.5.6.11.2	X			
Maintainability	3.17	4.7				X
Line frequency variation	3.16.11.1.2	4.5.6.11.2	X			
Variation in humidity	3.16.11.1.3	4.5.6.16	X			
Variations in temperature	3.16.11.1.4	4.5.6.16	X			
Control variations	3.16.11.1.5	4.5.6.11.5	X		X	
Frequency drift rate	3.16.11.1.6.2	4.5.6.11.3	X		X	
Phase stability	3.16.11.1.7	4.5.6.11.8	X		X	
Regeneration	3.16.12	4.5.6.12	X		X	
Oscillator radiation	3.16.13	4.5.6.13	X		X	
Frequency standard calibration	3.16.14	4.5.6.14	X		X	
Adjustment range	3.16.14.1.1	4.5.6.11.6	X		X	
External frequency standard	3.16.14.2	4.5.6.14.1	X		X	
Steady-state voltage	3.6	4.5.6.19	X		X	
Power and power factor	3.6	4.5.1	X		X	
Steady-state frequency	3.6.1	4.5.6.19	X		X	
Transient voltage	3.6.1	4.5.6.19	X		X	
Transient frequency	3.6.1	4.5.6.19	X		X	
Power interruption	3.6.1	4.5.6.20	X		X	
Temperature sequence	3.11	4.5.6.16	X			X
Low temperature	3.11	4.5.6.16	X			X
High temperature	3.11	4.5.6.16	X			X
Humidity	3.11	4.5.6.16	X			X
Salt fog (spray)	3.11	4.5.1	X			X
Accelerated life	3.16.17	4.5.6.24	X			
Reliability	3.13	4.6				
Enclosure	3.15.1.3	4.5.6.18	X			X
Shock	3.15.1.24.1	4.5.6.17	X			X
Vibration	3.15.1.24.2	4.5.6.17	X			X
Inclination	3.15.1.24.3	4.5.6.17	X			
Thermal design	3.15.1.37	4.5.1	X			
EMC	3.16.16	4.5.6.25	X			X
Workmanship screen	3.18.2	4.5.3	X	X		

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4.

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edges of the front panel. The mounting brackets and

4.4.1 Production inspection (Group A). Production inspection shall be conducted on each equipment offered for delivery. Production inspection shall encompass such examinations and tests which will prove the workmanship and reveal omissions and errors of the production process, such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls or adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examination and tests shown in Group A, of TABLE V, subject to the conditions of 4.4.1.1.

4.4.1.1 Group A tests. The Group A tests shall be performed in accordance with the test methods of 4.5.5 except as modified by a through d:

a. Maximum undistorted output: 2.1 MHz
b. Output signal-to-hum ratio: 2.1 MHz
c. AGC characteristics: 2.1 MHz
d. Frequency stability: 5 MHz

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis in accordance with the inspection procedures of MIL-STD-105, using inspection level S-3 with an acceptable quality level of 6.5 percent for each attribute. The maximum of sample test units shall be two each. Production control inspection shall consist of the examinations and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of the performance with other equipment in a system. These tests shall be performed on the complete equipment as offered for delivery. Production control inspection shall include the examinations and tests shown in Group B of TABLE V. Production control inspection shall be conducted on inspection lots that have passed the production inspection of 4.4.1. The maximum number of sample test units shall be two each.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from the new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be conducted on a sampling basis as specified in 4.4.3.1. Environmental inspection shall encompass environmental tests to prove the durability of materials, parts, units, and the equipment as a whole; life tests, simulated service tests; tests of the effects of shock, vibration, and inclination. Environmental inspection shall include the examinations and tests shown in Group C of TABLE V.

4.4.3.1 Sampling for inspection of equipment. The equipment selected for environmental inspection shall pass all tests, except electromagnetic interference (EMI), before the lot of equipments the sample is selected from can be moved from the contractor's facilities. One equipment from the first month's production lot, or from the first 50 units produced, shall be subjected to environmental inspection. One equipment from each successive production lot of 100, or one per month if less than 100 are produced monthly, shall be subjected to environmental inspection. One equipment from each 200 equipments produced shall be subjected to EMI tests. Environmental inspection shall be performed on equipments that have passed the production inspection of 4.4.1.

4.4.3.2 Noncompliance. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative (QAR) the results thereof and corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.6).

4.4.4 Reinspection of conforming production control and environmental inspection sample units. Unless otherwise specified (see 6.2.1), sample units which have been subjected to, and have passed, production control or environmental inspections, or both, may be accepted on the contract provided that they are resubjected to, and pass, production inspection after repair of all defects.

4.5 Test methods. Test methods shall be as specified in 4.5.1 through 4.5.5.

4.5.1 Examinations and tests. When 4.5.1 is specified in TABLE V, the test method specified in MIL-E-16400 shall be applied.

4.5.2 Workmanship. Workmanship shall be as specified in 4.5.2.1 through 4.5.3.2.

4.5.2.1 General workmanship and soldering. The equipment, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process for conformance to the requirements of 3.18.1. Each solder connection and associated wiring leads shall be visually examined.

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4.5.3 Workmanship screen. Prior to conducting temperature cycling, vibration shall be performed on each equipment. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration.

4.5.3.1 Random vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 1. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth in accordance with MIL-STD-810C, Method 514.2, Random vibration test paragraph, shall be used for control and analysis of the acceleration spectral density. The instantaneous acceleration peaks shall be limited to three times the acceleration rms value. All failures occurring during screening shall be corrected and the vibration resumed.

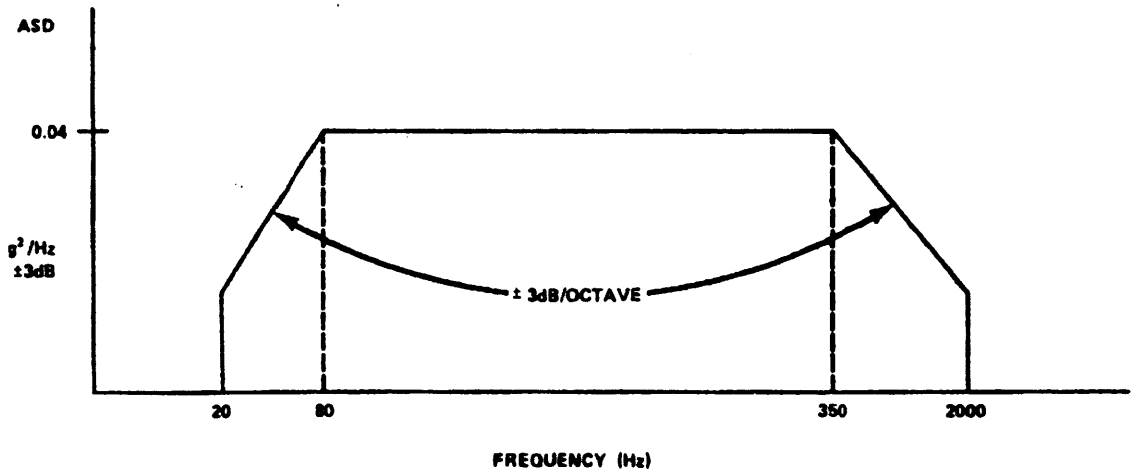


FIGURE 1. Random vibration curve.

4.5.3.2 Temperature cycling. Each unit shall be subjected to 10 cycles of the temperature curve shown in FIGURE 2. The temperature rate of change shall be not less than 5°C per minute. Unit power shall be turned on and off at the indicated times. The unit shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the unit reaches 80 percent of the chamber temperature. When failures occur, the unit shall be reworked and the cycling continued for a cumulative total of 10 cycles.

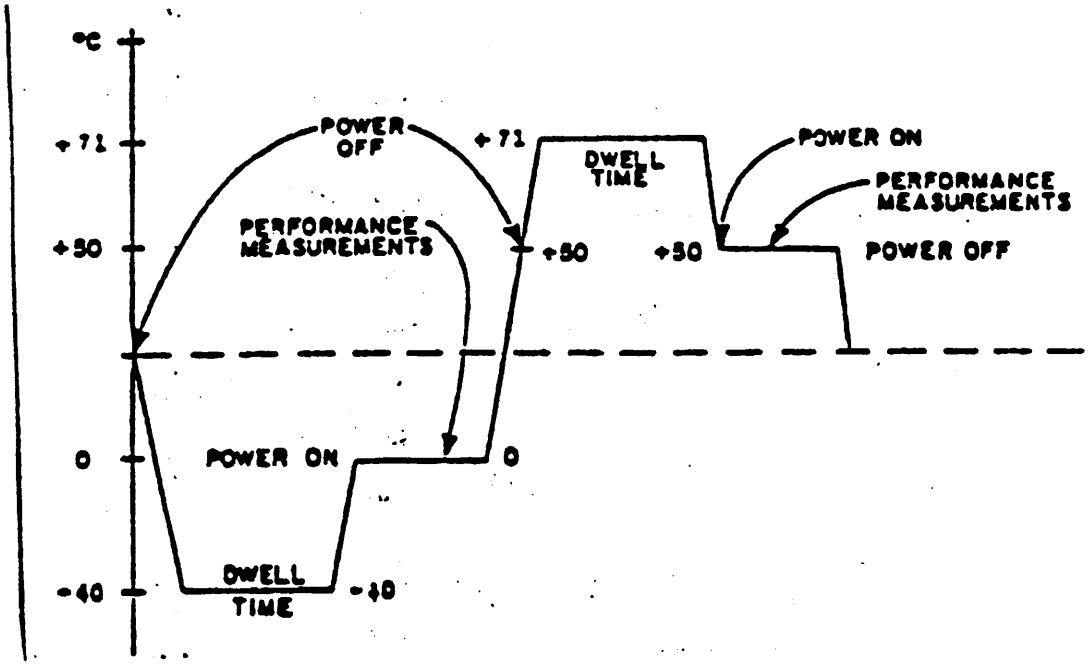


FIGURE 2. Temperature curve (one cycle) (not to scale).

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4.5.4 Leakage current test. The leakage current test shall be as specified in 4.5.4.1 and 4.5.4.2.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.4.1 Equipment connections. Each equipment directly connected to an external power source and units deriving from the equipment shall be placed on an insulated surface. All safety ground conductors between the equipment and units deriving power from the equipment shall be intact. The safety ground conductor between the equipment under test and the source power shall be opened during the test. The equipment shall be connected as shown in FIGURE 3 if it is connected to a single-phase source power, and as shown in FIGURE 4 if connected to three-phase source power.

4.5.4.2 Measurement. Leakage current shall be measured on the equipment in its normal operating configuration. Equipment controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the equipment is designed. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed and the voltage measured for every combination of switch positions available in FIGURE 3 and FIGURE 4. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.5 Cables. Interconnecting cables to be supplied with the end item shall be installed on the end item prior to any testing, and shall be used for the duration of all testing.

4.5.6 Detailed test methods. Detailed test methods shall be as specified in 4.5.6.1 through 4.5.6.25.

4.5.6.1 Sensitivity. Measurement shall be made at the 5 percent, 50 percent, and 95 percent points in each of the primary frequency bands for the USB mode, and at the frequency of poorest USB sensitivity for LSB, AM, and CW modes.

4.5.6.1.1 Gain. Measurement shall be made at the frequency of lowest gain as determined by the procedure of 4.5.6.1.2.

4.5.6.1.2 Gain variation measurement procedure. Apply a signal of one mV, open circuit, in series with 50 ohms, at a frequency of 2.1010 MHz through the equipment antenna input terminal. Tune the equipment to 2.1000 MHz in USB mode. Adjust the RF gain control for maximum and turn AGC to OFF. Adjust the audio line level control to produce 1 mW of audio output. Leaving the gain and level controls and the input signal amplitude fixed, tune the equipment and signal source together to produce a constant output frequency of 1 kHz. Within each MHz frequency band, note output for each setting of the 100 kHz selector, with all less significant selectors set at zero. Continue procedure with the 10 kHz and 1 kHz selector until maximum output is determined and recorded for each MHz band. Repeat the procedure, determining and recording minimum output in each band.

4.5.6.2 Selectivity (antenna input to detector input). Measurements shall be made in USB, LSB, and AM modes with the equipment tuned to 17.5500 MHz. AGC shall be inoperative and RF gain shall be maximum. Measurements shall be made of all parameters specified in 3.16.2.1.1 and 3.16.2.1.2, using an unmodulated test signal. Measurements shall be made using a tracking loop arrangement to provide a continuous X-Y plot of each sideband. The test signal amplitude shall be constant and shall be chosen to prevent signal path overload. Measurement system accuracy, resolution, and dynamic range shall be adequate to verify compliance with all specification requirements.

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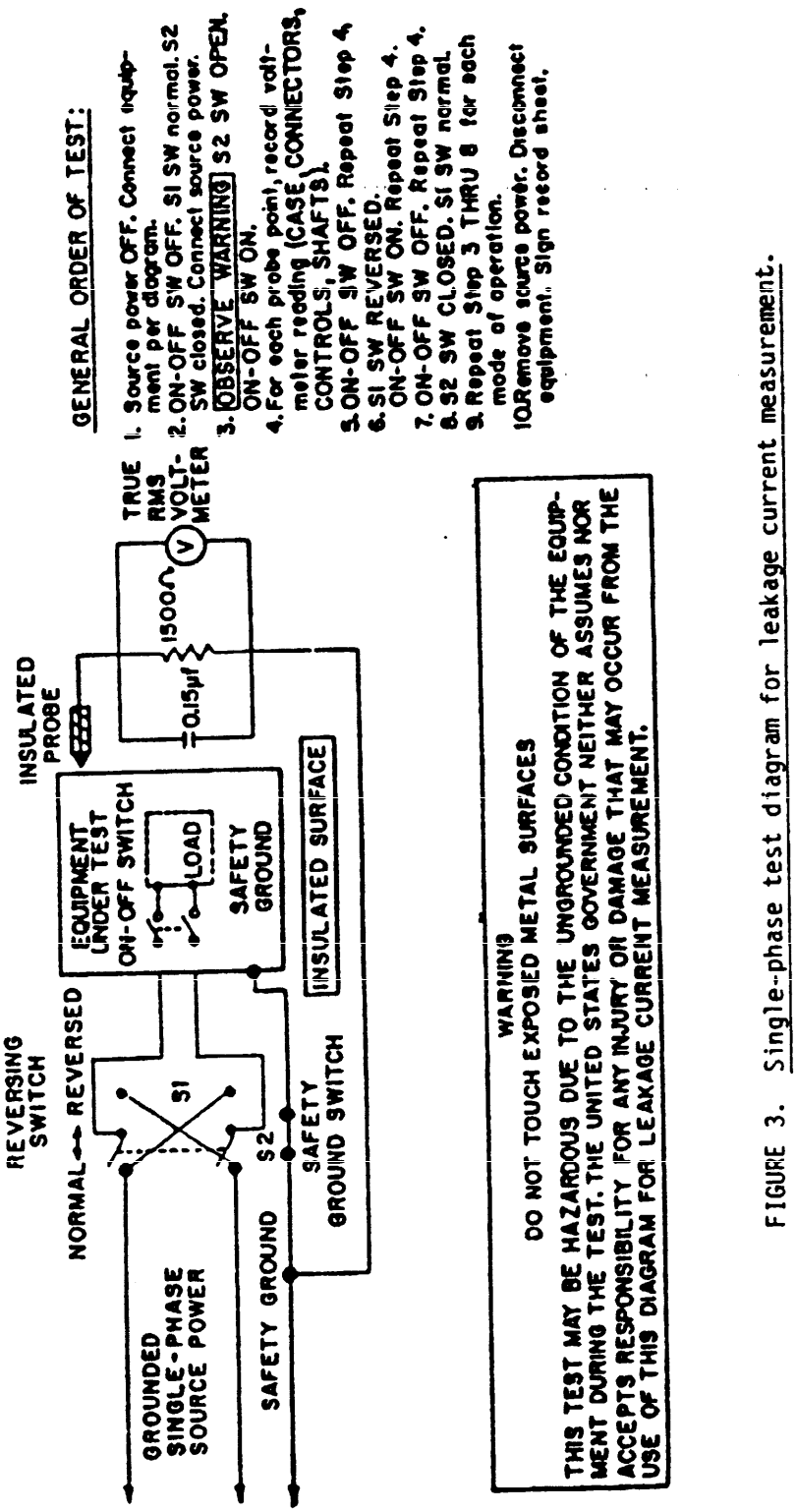
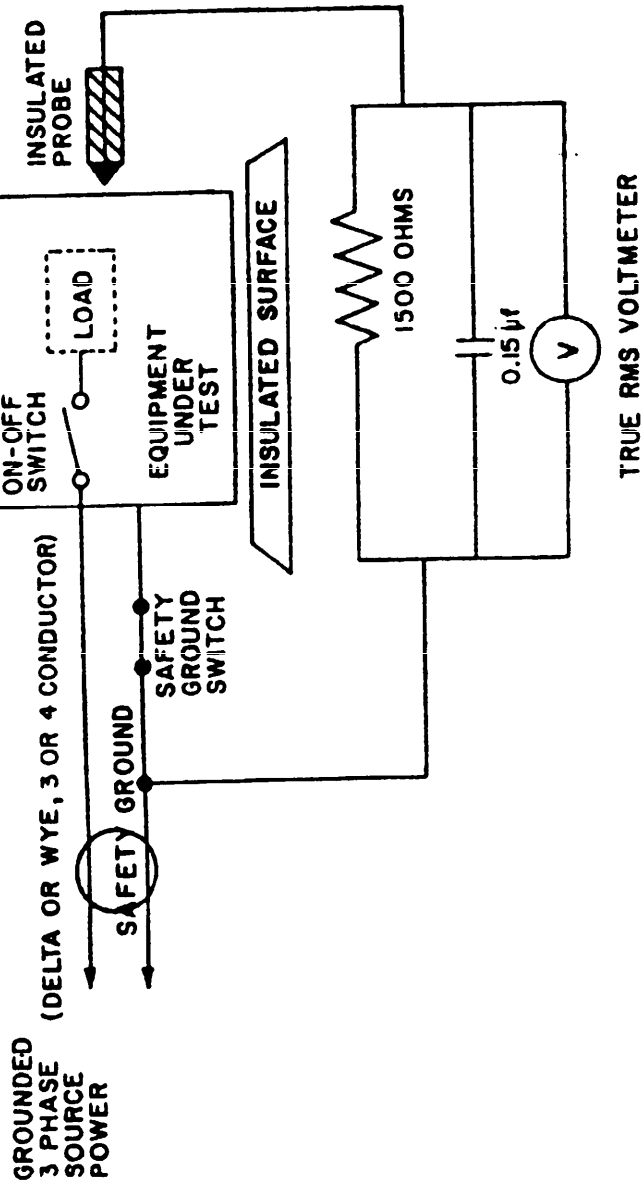


FIGURE 3. Single-phase test diagram for leakage current measurement.

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GENERAL ORDER OF TESTS:

1. Source power OFF. Connect equipment per diagram.
2. ON-OFF SW OFF. Safety ground switch closed. Connect source power.
3. OBSERVE WARNING. Safety ground switch OPEN. ON-OFF SW ON.
4. For each probe point, record voltmeter reading (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. Repeat Step 4.
6. Safety ground switch closed.
7. Repeat Step 3 THRU 6 for each mode of operation.
8. Remove source power. Disconnect equipment. Sign record sheet.

NOTE: 1. All three phases shall be connected during measurement.
2. The safety ground conductor shall not carry load current.

WARNING
DO NOT TOUCH EXPOSED METAL SURFACES
THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

FIGURE 4. Three-phase test diagram for leakage current measurement.

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4.5.6.2.1 Overall passband (antenna input to audio output). Measurements shall be made to USB and LSB modes with the equipment tuned to 17.5500 MHz. AGC shall be inoperative. Measurements shall be made using a tracking loop arrangement to provide a continuous X-Y plot of each sideband. The test signal amplitude shall be constant and shall be chosen to prevent signal path overload. Measurement system accuracy, resolution, and dynamic range shall be adequate to verify compliance with all specification requirements.

4.5.6.2.2 Overload selectivity (desensitization). Overload selectivity (desensitization) shall be measured with the equipment operating in the SSB mode, and with AGC off. An unmodulated signal shall be applied at resonance to the input of the equipment from a 50-ohm standard dummy antenna and adjusted to yield standard sensitivity. An unmodulated interfering signal shall be applied from the same standard dummy antenna at a frequency differing from the original signal by the IF bandwidth; the signal level shall be increased to the point where the S+N/N ratio of the original signal is decreased by 3 dB. The procedure shall be repeated for increasing frequency spacing of the desensitizing signal from the equipment operating frequency including a minimum number of points specified in 3.16.2.3.

4.5.6.2.3 Gain depression. At each point on the overload selectivity curve, the amount of gain depression for the desired signal shall be measured at the 3-dB desensitization point by means of a wave analyzer tuned to the desired audio output frequency.

4.5.6.2.4 Envelope delay distortion. Measurements shall be made in USB and LSB modes at any suitable radio operating frequency. An envelope delay test set shall be used to generate an audio carrier which shall be swept from 300 Hz to 3100 Hz. The audio carrier shall be 50 percent amplitude modulated at 83.3 Hz, and the resulting signal shall be translated to the equipment operating frequency and applied to the equipment antenna input terminals. The equipment USB and LSB audio outputs shall be returned (sequentially) to the envelope delay test set receive terminals to permit measurement of relative envelope delay for each sideband. An XY plotter shall be used to provide continuous record of envelope delay as a function of frequency for each sideband.

4.5.6.3 Spurious responses. Each primary frequency band shall be completely explored for evidence of the types of spurious responses specified in a through e which fall outside of the allowable limits as specified in 3.16.3:

a. Internally generated spurious outputs (AGC off)
b. Image rejection. Measurements shall be made at the nominal end frequencies of each primary band, for first, second, and third mixer images.
c. IF rejection. Measurements shall be made at the nominal end frequencies of each primary band.
d. Other single-signal spurious responses
e. Front-end intermodulation (2.4 MHz, 17.1 MHz, and 26.1 MHz)

4.5.6.3.1 Cross modulation selectivity. Cross modulation selectivity shall be measured with AGC on (slow), by applying a 100 µV signal that is modulated 90 percent at 1000 Hz at the resonant frequency to which the equipment is tuned, and adjusting the gain of the set to produce 60 mW of AF output (across the line output connector). The modulation of the desired signal shall then be removed and an interfering off-resonant signal, modulated 90 percent at 1000 Hz, shall be applied and adjusted in frequency and amplitude to produce a cross modulation AF output level of 6 mW without any change in the control settings of the equipment. This interfering signal shall then be readjusted, in both frequency and amplitude, on both sides of the resonant frequency to which the equipment is tuned, so as to produce 6 mW of interfering output as the result of cross modulation at the various frequencies of resonance. The interfering signal input level shall range from a value which begins to produce the cross modulation phenomenon at the closest possible spacing to the desired signal frequency, up to interfering signal levels of at least 5 V. The cross modulation selectivity shall not be degraded for desired signal inputs up to at least 60 dB above standard sensitivity input. The cross modulation performance shall be measured in the A3j mode. Cross modulation selectivity shall be measured at four points; that is f_o ±15 percent, and either f_o ±2.5 percent or f_o ±4 percent, as appropriate. Measurement shall be made for operating frequencies of 2.1 MHz, 24.9 MHz, and 29.9 MHz.

4.5.6.3.2 Internal spurious responses. Internally generated spurious responses shall be measured in USB and LSB modes for equipment operation from both internal and external frequency standards. Initial measurements shall be made in the USB mode with equipment operation from the internal frequency standard. The equipment antenna input terminals shall be connected to a shielded

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50-ohm resistor and the AGC turned off. The equipment shall be tuned to 2.990 MHz and the USB audio output signal monitored with an audio spectrum analyzer and earphones. The equipment shall be tuned throughout the complete frequency range in increments of no more than 100 kHz with dial settings of xx.xx90 MHz. Spurious responses detected shall be evaluated against the requirements of 3.16.3.1.1 by substitution of a standard signal generator for the 50-ohm input termination, offsetting the generator frequency to permit amplitude comparison of the internal spurious response and that due to the applied signal, using the audio spectrum analyzer. Measurements shall be repeated for LSB operation, using dial setting of xx.xx10 MHz.

4.5.6.4 Audio output measurements. Audio output measurements shall be as specified in 4.5.6.4.1 through 4.5.6.4.5.

4.5.6.4.1 Audio output voltage regulation. Measurements of audio output variation with change of load impedance and with variation of input signal level shall be made at the line output of the audio amplifier in USB and LSB. Tests shall be performed to ensure that no damage to output transistors occurs when operated into short and open circuit loads under zero drive and maximum drive conditions.

4.5.6.4.2 Maximum undistorted output. Measurement shall be made in the AM, USB, and LSB modes at a nominal RF input frequency of 2.1 MHz. The AGC shall be operative, and measurements shall be made for the FAST and SLOW positions of the AGC time constant selector switch. Measurments shall be made on both rear-mounted line output receptacles and panel-mounted head telephone jacks. AF output data shall be obtained for applied RF signal inputs from 1000 µV up to 1 V, with input level recorded in decade steps. Measurements shall be made at audio frequencies of 300 Hz, 1000 Hz, and 3500 Hz.

4.5.6.4.3 Range and linearity of audio level controls. Measurements shall be made of line output and head telephone control characteristics in the USB mode. AGC shall be active in the SLOW position of the AGC switch. Audio output as a function of angular rotation of the audio gain controls in increments of 10 degrees or less shall be recorded starting from the fully counterclockwise position. The measurements shall be performed at nominal operating frequency of 2.1000 MHz.

4.5.6.4.4 Audio level meter. Measurements shall be made to verify compliance with 3.16.4.3.1.

4.5.6.4.5 Output hum and noise. Measurements shall be made for AM, USB, and LSB modes at RF input frequencies of 2.1 MHz, 17.1 MHz, and 29.9 MHz. The input signal level shall be 1000 µV, supplied from a standard 50-ohm dummy antenna. For SSB modes, the signal shall be unmodulated; for AM, the signal shall be modulated 90 percent at 1000 Hz. Measurements shall be made for power line frequencies of 60 Hz and 400 Hz, as specified in a and b:

a. Set equipment RF gain to maximum and adjust line level control to produce standard output of 6 mW. With this level as a reference, measure relative amplitude of power line related components in the output, using a wave analyzer.

b. Set equipment line control to maximum gain, and adjust the manual RF gain control to produce standard output of 6 mW. With this level as a reference, remove the RF input signal and measure the relative amplitude of the remaining hum and noise components with a broadband voltmeter.

4.5.6.5 Manual RF gain control. Range and linearity measurement shall be made with the equipment conditioned for USB (AGC inoperative) for an applied nominal RF input frequency of 2.1 MHz. Record gain reduction in dB as a function of angular rotation in intervals of not greater than 10 degrees.

4.5.6.6 AGC characteristics. Equipment AGC amplitude characteristics and time constants shall be measured for USB, LSB, and AM operation at the equipment low gain frequency. AGC linearity shall be verified. Measurements shall be made of maximum change in audio output level for RF inputs between that which produces standard output S+N/N in the SSB modes and 1 V, open circuit. No evidence of blocking or saturation of output signal shall occur.

4.5.6.6.1 AGC time constants. Separate determination of each of the AGC time constants shall be made by switching the RF input amplitude between the level which produces standard output response and various large signal levels from 100 µV to 0.1 V, open circuit, and observing the audio output. Oscilloscope photographs of the equipment audio output shall be obtained for attack and release characteristics, triggering the oscilloscope sweep on new RF amplitude transition.

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4.5.6.6.2 Resonant overload. Measurement shall be made in the USB, LSB, and AM (90 percent modulation) modes at the equipment high gain frequency. The equipment AGC shall be disabled, and AF and RF level controls shall be set for maximum gain.

4.5.6.6.3 Time of recovery from surge inputs. Measurement shall be made in USB and LSB modes at the equipment high gain frequency.

4.5.6.7 BFO control. The output frequency range of the BFO control shall be measured at an RF input frequency of 2.1 MHz.

4.5.6.8 In-band intermodulation. Measurements shall be made at equipment high and low gain frequencies, in USB and LSB modes, and for FAST and SLOW AGC modes. RF signal frequencies shall be such that audio output tone frequencies are 935 Hz and 1045 Hz. Measurements shall be made with RF signal amplitudes of -37 dBm and +1 dBm per tone, and audio output of 3 Vrms per tone.

4.5.6.9 RF input circuit. Threshold of operation of the antenna input protection circuit and insertion loss in the operated condition shall be measured at the mid-frequency in each primary (MHz) band to show compliance with the requirements of 3.16.9. Equipment input circuit protection shall be verified in each primary (MHz) band. Equipment input circuit protection shall be verified by keying drive to a suitable RF power amplifier held in a key-down condition. The power amplifier shall provide not less than 100 V open circuit in series with 50 ohms, and the protection circuit shall be operated a minimum of 10 times at each test frequency. Circuit operating threshold shall be measured at the center of each MHz band. Desensitization produced by operation of the circuit by an out-of-band transmitter shall be measured at the center of each MHz band, simulating an overload by opening the normal connection to antenna overload board terminal Z1A9E6, substituting a ground, and then measuring the change in sensitivity.

4.5.6.10 Accuracy of main tuning indicator calibration. Measurement shall be made at frequencies required to verify proper operation of each frequency selector in each digital position.

4.5.6.10.1 Tuning time. Measurement shall be made of the time required to switch from 2.0000 MHz to 29.9999 MHz.

4.5.6.10.2 Frequency vernier range. Measurements shall be made to verify compliance with 3.16.10.2.

4.5.6.11 Frequency and gain stability. Frequency and gain stability shall be measured in accordance with the conditions specified in 4.5.6.11.1 through 4.5.6.11.7.

4.5.6.11.1 Sixty-minute warm-up period. The warm-up period shall begin after exposure of the equipment to the ambient temperature for a period of not less than 6 hours with no power applied, to establish the temperature at the same value as the ambient. Frequency measurements shall be made at intervals of not more than 2 minutes for the first 12 minutes, with the first measurements made 2 minutes after power is applied. Measurements shall be made at intervals of not more than 5 minutes for the next 18 minutes, and at intervals of not more than 10 minutes for the last 30 minutes.

4.5.6.11.2 Line voltage frequency variation. Measurements shall be made in accordance with the procedures specified in MIL-E-16400. A minimum stabilization period of 30 minutes shall be observed following each change of supply voltage and frequency.

4.5.6.11.3 Twenty-four-hour frequency drifts. All equipment power shall be removed for a period of at least 6 hours prior to stability measurements. The equipment shall be energized for a stabilization period not to exceed 48 hours, and operated during the test period with temperature, humidity, and line voltage controlled as specified in 3.16.11.1.6.2.

4.5.6.11.4 Twenty-four-hour gain stability. Gain stability shall be measured in the ISB mode (signal in both sidebands), with AGC off. Equipment conditioning periods, and temperature, humidity, and line voltage control, shall be as specified in 4.5.6.11.3.

4.5.6.11.5 Control variation. Measurement shall be made to verify compliance with 3.16.11.1.5.

4.5.6.11.6 Frequency standard adjustment range. Measurement shall be made to verify compliance with 3.16.14.1.1.

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4.5.6.11.7 Vernier stability. Measurement shall be made to verify compliance with 3.16.10.2.

4.5.6.11.8 Phase stability. Overall phase stability of the equipment shall be measured to verify compliance with 3.16.11.1.7. A phase-stable RF signal shall be applied to the equipment antenna terminals, and adjusted in frequency to produce a suitable audio frequency output. Measurements shall be made of the average phase deviation in the audio output signal in measurement periods of 9.09 ms, over the number of measurement periods necessary to establish the specified probability with a confidence of at least 95 percent. Measurements shall be made at the test frequencies specified in 4.5.6.10.

4.5.6.12 Regeneration. Each primary frequency band shall be carefully explored for evidence of regeneration.

4.5.6.13 Oscillator radiation. Measurements shall be made at the nominal end frequencies of each primary MHz band.

4.5.6.14 Frequency standard calibration. Measurements shall be made to determine compliance with the requirements of 3.16.14 through 3.16.14.3.

4.5.6.14.1 External frequency standard. Operation using an external frequency standard shall be made to verify compliance with 3.16.14.2.

4.5.6.15 Frequency standard construction. Frequency standard assemblies shall be inspected to verify compliance with 3.15.1.36.

4.5.6.16 Temperature and humidity. Tests shall be conducted in accordance with MIL-E-16400. Measurements of equipment gain variation (AGC off) and frequency stability shall be made to determine compliance with the requirements of 3.16.11.1.3 and 3.16.11.1.4.2. Measurements shall be made in USB, LSB, and CW modes for a minimum of one high band and one low band frequency. Frequency accuracy (synthesizer lock) checks, phase jitter, and envelope delay measurements shall be made during temperature operating test at the end of the period of operation at 0°C, after 4 hours of operation with the equipment stabilized at +50°C, and after return to room temperature. During the last hour of hign temperature operating tests, air flow produced by forced convection in the test chamber shall be eliminated. During this period, air temperature in the immediate vicinity of the equipment under test shall be controlled to 50°C by radiation-baffled heating and cooling sources as required. This test is intended to simulate the worst-case shipboard thermal environment (see 4.5.6.16.1). During humidity tests, measurements shall be made at the end of the second and fifth cycles (see 3.16.10.1 and 4.5.6.10).

4.5.6.16.1 Chamber temperature control provisions. The sources for maintaining the chamber environment shall be arranged or suitably baffled so that no surface at a temperature other than the desired ambient is visible to the unit under test. The maximum allowable temperature difference between ary two points in the chamber shall be 10°C. The chamber control provisions shall be capable of maintaining the temperature indicated by a selected chamber ambient sensor constant within ±1°C of any selected test temperature.

4.5.6.17 Shock, vibration, and inclination. Prior to and after completion of shock, vibration, and inclination tests, the tests specified in a through c shall be performed. The shock, vibration, and inclination tests shall be performed to verify compliance with 3.15.1.24.1 through 3.15.1.24.3.

a. Sensitivity measurements in accordance with 4.5.6.1 in the A1 USB, and LSB modes for the 2 MHz, 17 MHz, and 29 Hz bands only
b. Synthesizer lock checks in accordance with 4.5.6.10.
c. Intermodulation distortion in accordance with 4.5.6.8. During vibration tests equipment operation and monitoring shall be in accordance with requirements of 3.16.8.1.

4.5.6.18 Enclosure. Tests shall be conducted in accordance with the Enclosure tests paragraph of MIL-E-16400. Measurements of gain and frequency stability shall be made prior to and after testing. Measurements shall be within the limits established in 3.16.11.1.

4.5.6.19 Supply line voltage and frequency variation and transients. Tests shall be conducted in accordance with MIL-E-16400. Reference gain, sensitivity, and frequency measurements shall be made prior to start of testing. Variation of frequency and gain with line voltage and frequency shall be measured to determine compliance with the requirements of 3.16.11.1.2 for nominal line voltage of 115 VAC and for nominal line frequencies of 50 Hz, 60 Hz, and 400 Hz. Following performance of the tests specified in MIL-E-16400, no permanent changes shall be produced in frequency, gain, or sensitivity.

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4.5.6.20 Power interruption and spike tests. Power interruption tests shall be conducted in accordance with the Power interruption test paragraph of MIL-E-16400. The 2500 V spike test shall be performed in accordance with the requirements of DoD-STD-1399, Section 300. Source impedance for the spike shall be 25 ohms, resistive. Spikes shall be delivered line-line and line-ground for each primary power line. For each test, a positive-going spike shall be applied at the positive crest of the supply line voltage waveform and at the zero axis crossing, and a negative-going spike shall be applied at the negative crest.

4.5.6.21 Chassis orientation. Verification of the ability to orient the chassis ±90 degrees shall be performed.

4.5.6.22 Mode selector switch. Mode selector switch functions specified in 3.15.1.8.1 shall be verified.

4.5.6.23 Assembly identification. All applicable assemblies shall be inspected to verify compliance with 3.15.1.18.

4.5.6.24 Accelerated life. Measurements of sensitivity (2 MHz, 17 MHz, 29 MHz bands) and accuracy of main tuning indicator calibration (synthesizer lock) shall be made in accordance with 4.5.6.1 and 4.5.6.10.

4.5.6.25 EMC. Compliance with 3.16.16 shall be verified by tests performed in accordance with MIL-STD-462. Group C inspection shall be performed for tests CE03, CS06, and RE02 and any other tests requested by the Navy for this procurement (see 6.2.2).

4.6 Reliability qualification test. A reliability qualification test should be performed in accordance with Test Plan IVC of MIL-STD-781 on at least two units from the first month's production. Environmental conditions and operational duty cycles for reliability qualification testing shall be as specified herein.

4.6.1 Failure. After the initial start of the test, any parameter deviation which requires adjustment or alignment of the equipment under test shall be classified as a failure (see 6.7.13).

4.6.2 Environmental conditions for reliability testing. The environmental conditions specified in a through c shall be maintained throughout the reliability qualification test:

a. Ambient temperature shall be +50°C, +0, -30°C
b. Vibration shall be 1G ±10 percent peak acceleration value at any nonresonant frequency between 20 Hz and 60 Hz measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.
c. Input voltage shall be 115 VAC, 60 Hz or 400 Hz. Input voltage shall be cycled in accordance with the three-day cycle specified in 1 through 3:
   1. Nominal
   2. Nominal +10 percent
   3. Nominal -10 percent

4.6.3 Performance tests. At least daily, performance measurements, shall be made to verify the performance characteristics specified in a through c:

a. Gain and sensitivity: Equipment sensitivity and gain shall conform to the requirements of 3.16.1.1 and 3.16.1.2 (SSB measurements only).
b. Synthesizer lock: The synthesizer shall remain locked to the internal frequency standard at all times.
c. Frequency stability: The drift of the internal frequency standard shall be not greater than 3 parts in 10^8 per day, nor 3 parts in 10^7 per month.

4.6.4 Test frequencies. During the reliability tests, at least one frequency shall be tested in each band as specified in a through d:

a. F1 = 2.0015 MHz
b. F2 = 5.3340 MHz
c. F3 = 10.6655 MHz
d. F4 = 29.9990 MHz

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4.6.5 Test schedule. The test schedule of TABLE VI, including the input line voltage schedule of 4.6.2, shall be conducted. The test shall consist of three 8-hour schedules per day, the first being manned during which performance measurements shall be made. The second two may be unmanned and pertinent data shall be automatically recorded only to the extent that instrumentation permits. Operating schedules shall be as specified in TABLE VI.

TABLE VI. Test schedule.

Manned schedule	Unmanned schedule
0-2 hours equipment ON 2-3 hours performance measurements 3-4 hours equipment OFF 4-6 hours equipment ON 6-7 hours equipment measurements 7-8 hours equipment OFF	0-3 hours equipment ON 3-4 hours equipment OFF 4-7 hours equipment ON 7-8 hours equipment OFF

4.6.5.1 Manned operation schedule. The mode and frequency of operation for each of the operational ON periods of the manned schedule shall occur in the sequence specified in TABLE VII. During each period, an appropriate RF signal shall be used to simulate actual equipment operation in the particular operating mode under test.

TABLE VII. Operating schedules.

Day 1/	Frequency	Mode of operation
1	F1 first 4 hours of manned schedule F2 remainder of the day	CW(A1) keyed at a rate of 32 baud
2	F3 first 4 hours of manned schedule F4 remainder of the day	AM(A3h) Carrier modulate 30-50 percent at 1000 Hz
3	F1 first 4 hours of manned schedule F2 remainder of the day	ISB(A3b) single-tone modulation per sideband
4	F3 first 4 hours of manned schedule F4 remainder of the day	ON-OFF (5 Hz rate) signal, 1000 Hz single tone in each sideband, ISB mode, FAST AGC

1/ The same sequence shall be used on the next four days, except that the sequence of frequencies shall be reversed, F4 and F3 on the fifth day, F2 and F1 on the sixth day, F4 and F3 on the seventh day, and F2 and F1 on the eighth day. The schedule specified in TABLE VII shall be repeated as necessary.

4.6.5.2 Unmanned operating schedule. The mode of operation during the unmanned periods shall be SSB (A3j). An unmodulated RF signal shall be applied to the antenna port to produce a 1-kHz audio output. The test frequency to be used during each unmanned period shall be the lower frequency utilized during the previous manned period (see 4.6.5.1).

4.7 Maintainability demonstration. Maintainability demonstration shall be as specified in 4.7.1 through 4.7.7.

4.7.1 Maintainability equipment demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be used to verify that the equipment MTTR requirement is fulfilled. The maintainability demonstration shall be performed just once, during the first 3 month's production.

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4.7.1.1 Demonstration conditions. The equipment maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual PCBs) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. One hundred candidate corrective maintenance faults shall be made available for each candidate task as specified in a through c:

a. Designation of specification faulty part
b. Failure mode
c. Means of introducing fault (substitution of faulty part or simulation thereof)

The procuring activity or its authorized representative, shall use the candidate tasks as a guide to determine a sample of 50 tasks for the demonstration.

4.7.2 Accept or reject criteria. Accept or reject criteria shall be as specified in 4.7.2.1.

4.7.2.1 Corrective maintenance. The accept or reject criteria for the demonstration of the corrective maintenance times for the demonstration shall be as specified in TABLE VIII.

TABLE VIII. Acceptance criteria.

	MTTR	M_{maxct}
Acceptance level	20	0
Sample size	50	50

Acceptance shall occur when the number of observed corrective maintenance task times which exceed the required value of each specified index, MTTR or M_{maxct}, is less than or equal to that specified in TABLE VIII corresponding to each index for the specified confidence level. The duration of each task shall be compared to the required value(s) and determined as greater than or less than each index. The actual value shall also be determined. An accept decision can only be made when an accept decision is made for both MTTR and M_{maxct}. Accept or reject criteria for preventive maintenance shall be the capability to perform each preventive maintenance task with no degradation of system performance. If degradation of system operations exists, the preventive maintenance test will have resulted in a reject decision. More than one erroneous failure indication shall constitute failure of the demonstration.

4.7.3 Demonstration environment. The demonstration at the organizational level shall be performed in an environment similar to that in which the equipment will be installed.

4.7.4 Technicians. The Government reserves the right to provide and select technicians to perform the maintainability demonstration.

4.7.5 Technical documentation. Technical documentation shall be limited to the technical manual and related maintenance documentation delivered with the equipment.

4.7.6 Rejection. Failure to conform to any of the requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the contracting officer shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies and
b. Upon approval of an approach, repeat the demonstration until an accept decision is reached.

4.7.7 Acceptance. The maintainability demonstration shall demonstrate that the equipment conforms to the accept criteria prior to delivery.

4.8 Inspection of preparation for delivery. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only to direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

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5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for general purpose voice, CW, radio teletypewriter, and data communications, and may be used for surface ship, submarine, and fixed shore installation.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 3.1.1 and 4.3
c. GFE, if required (see 3.2)
d. Interchangeability requirements, if required (see 3.3)
e. Quantity of accessory items required (see 3.4.1)
   1. Shock and vibration mounting assembly (see 3.8.1)
   2. Extender cables (see 3.15.1.35 and 3.15.1.35.1)
   3. Extender cards (see 3.15.1.35 and 3.15.1.35.2)
f. Rack mounting brackets, as required (see 3.15.1.2.2)
g. When reinspected production control and environmental sample units may not be accepted (see 4.4.4)
h. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List, incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.5.6.25	Electromagnetic Interference Control Plan	DI-R-7061
4.5.6.25	Electromagnetic Interference Test Report	DI-R-7062
4.5.6.25	Electromagnetic Interference Test Plan	DI-R-7063

(DIDs related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item(s) should be a first article sample. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 GFE. The contracting officer should arrange to furnish the equipment specified in 3.2.

6.5 Nomenclature. Nomenclature referred to herein assigned to the GFE units may be subject to change on any contract resulting in equipment produced in accordance with this specification.

6.6 Environmental inspection. Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

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6.7 Definitions. Definitions of terms used in this specification are given in 6.7.1 through 6.7.13.

6.7.1 Standard dummy antenna. The standard dummy antenna is a device whose electrical constants are such as to simulate the impedance characteristics of antennas suitable for use with each equipment throughout the frequency range.

6.7.1.1 Dummy antenna characteristics. A device conforming to the requirement of 6.7.1 for this equipment is a 50-ohm nonreactive resistor. This resistance includes the output resistance of the standard signal generator. If necessary, resistance should be added externally to the standard signal generator to produce a total of 50 ohms when added to the effective output resistance of the standard signal generator.

6.7.1.2 Application and intent. The intent of 6.7.1.1 is that the standard signal generator produces the equivalent of the signal voltage induced by an electromagnetic field in an antenna which has the constants that are simulated by the dummy antenna. Hence, the voltage reading obtained from the standard signal generator is the open circuit voltage in series with the dummy antenna. This definition applies to all voltages specified relative to the equipment antenna input terminals. It should be noted that many standard signal generators do not indicate open circuit voltage, but rather the voltage which appears across a matched load. For these generators, the desired open circuit voltage is equal to twice the indicated output.

6.7.2 Standard output S+N/N ratio. Standard output S+N/N is 10 dB. This ratio should be determined in terms of rms values.

6.7.2.1 A1 and sideband reception modes. For A1 and sideband reception modes, the output S+N/N ratio is the ratio between the audio output resulting from an unmodulated RF input signal producing the standard beat note frequency and the output due to noise when the RF input signal is removed.

6.7.2.2 A2 and A3h modes of reception. For A2 and A3h modes of reception, the output S+N/N ratio is the ratio between the audio output resulting from the RF input signal modulated 30 percent at the standard modulation frequency and the output due to noise when the modulation is removed from the RF carrier.

6.7.3 Standard audio output. Standard audio output is a nominal 6 mW dissipated into a 600-ohm nonreactive resistance across the AF line output terminals.

6.7.4 Standard gain. Standard gain is that gain which produces standard output S+N/N ratio (see 6.7.2) with standard audio output.

6.7.5 Standard modulation frequency. Standard modulation frequency is 1000 Hz.

6.7.6 Standard beat note frequency. Standard beat note frequency for CW and SSB reception is 1000 Hz.

6.7.7 Sensitivity. Sensitivity of the equipment is that characteristic which determines the minimum strength of an input signal capable of producing a desired output S+N/N ratio.

6.7.7.1 Standard CW sensitivity. Standard CW (A1) sensitivity is the number of µV (open circuit) of unmodulated signal from a 50-ohm standard dummy antenna required to produce a standard output S+N/N ratio (see 6.7.2.1).

6.7.7.2 Standard SSB sensitivity. Standard SSB (A3j) sensitivity is the number of µV of unmodulated signal from a 50-ohm standard dummy antenna required to produce a standard output S+N/N ratio (see 6.7.2.1).

6.7.7.3 Standard AM sensitivity. Standard AM (A3h) sensitivity is the number of µV of signal from a 50-ohm standard dummy antenna, modulated 30 percent, required to produce a standard output S+N/N ratio (see 6.7.2.2).

6.7.8 Selectivity. Selectivity of the equipment is that characteristic which determines the extent to which it is capable of discriminating between a desired RF signal and any undesired signals at frequencies outside the bandwidth required by the desired signal information spectrum.

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6.7.8.1 Antenna input to detector input. Equipment RF and IF selectivity determined primarily by the degree to which the information band filters are capable of rejecting undesired signals at frequencies immediately adjacent to the desired signal information spectrum. Rejection of undesired signals is expressed in terms of dB with respect to the peak of the desired signal passband ripple.

6.7.8.2 Overall passband (antenna input to audio input). Selectivity determined by the combined effects of all frequency selective circuits in the entire signal path.

6.7.8.3 Overload selectivity. Overload selectivity (desensitization) is the degree to which the RF amplifiers of the equipment are capable of rejecting undesired signals at frequencies outside the passband of the RF circuitry. Protection of the equipment front end signal path circuits is required to avoid any decrease in a desired signal S+N/N ratio by a strong undesired signal.

6.7.9 Spurious responses. In general, a spurious response in an equipment is a presence of any signal in the passband of the IF amplifiers which is not a function of the desired signal alone. There are several specific classes of spurious response:

a. Internally generated spurious outputs caused solely by the presence of the oscillators inherent to the equipment
b. Signals which require the presence of a single strong signal external to the equipment. Specific types are IF and image responses.
c. Signals resulting from the intermodulation of two strong external signals in the first stages of the receiver prior to the IF amplifier
d. The super-imposition of modulation from a single strong signal external to the equipment on a desired signal by a cross modulation process

6.7.9.1 Image rejection. Images are the undesired second order products of each conversion, resulting from a signal on the opposite side of the injection frequency from the desired response. Image rejection is expressed as the ratio of input signal amplitudes, in dB at the image and resonant frequencies, for standard output S+N/N ratio.

6.7.9.2 IF rejection. IF rejection is the degree to which the equipment is capable of rejecting signals having the frequency(s) of the IF amplifier(s). IF rejection is expressed as the ratio of input signal amplitudes, in dB, at the IF and resonant frequencies, for standard output S+N/N ratio.

6.7.10 Resonant overload. Resonant overload output is the output at which the overall gain (with NO AGC) of the equipment, as conditioned for standard CW sensitivity, falls 3 dB with an increase of input carrier level.

6.7.11 Maximum undistorted output. Maximum undistorted power output is the output power level at which the total output harmonic content is 2 percent, in voltage, for an input carrier adjusted for the standard sideband measurement frequency (see 6.7.12) at the level of 1000 µV.

6.7.11.1 Total harmonic content. The total harmonic content is determined by measuring the residual voltages with the distortion analyzer after eliminating the fundamental of a sinusoidal modulating frequency.

6.7.12 Standard sideband measurement frequency. Standard sideband measurement frequency is 1000 Hz higher than the carrier for the USB and 1000 Hz lower than the carrier for the LSB.

6.7.13 Failure. Failure is defined as any malfunction or parameter deviation that prevents equipment from performing within the specified operational requirements.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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